

# Reliable Fuzzy Control for Active Suspension Systems With Actuator Delay and Fault

Hongyi Li, *Student Member, IEEE*, Honghai Liu, *Senior Member, IEEE*, Huijun Gao, *Senior Member, IEEE*, and Peng Shi, *Senior Member, IEEE*

**Abstract**—This paper is focused on reliable fuzzy  $H_\infty$  controller design for active suspension systems with actuator delay and fault. The Takagi–Sugeno (T–S) fuzzy model approach is adapted in this study with the consideration of the sprung and the unsprung mass variation, the actuator delay and fault, and other suspension performances. By the utilization of the parallel-distributed compensation scheme, a reliable fuzzy  $H_\infty$  performance analysis criterion is derived for the proposed T–S fuzzy model. Then, a reliable fuzzy  $H_\infty$  controller is designed such that the resulting T–S fuzzy system is reliable in the sense that it is asymptotically stable and has the prescribed  $H_\infty$  performance under given constraints. The existence condition of the reliable fuzzy  $H_\infty$  controller is obtained in terms of linear matrix inequalities (LMIs). Finally, a quarter-vehicle suspension model is used to demonstrate the effectiveness and potential of the proposed design techniques.

**Index Terms**—Active suspension systems with uncertainty, actuator delay, actuator fault, fuzzy control,  $H_\infty$  control.

## I. INTRODUCTION

VEHICLE engineering has seen the crucial role played by a vehicle suspension system in the evaluation of the vehicle dynamics performance. A suspension component has vital functions, e.g., to support the vehicle weight, to provide effective isolation of the chassis from road excitations, to keep tire contact with the ground, and to maintain the wheels at appropriate positions on the road surface. The roles of a vehicle suspension

system are to adequately guarantee the stability of the vehicle while providing as much comfort as possible for the passengers by serving the basic function of isolating passengers from road-induced vibration and shocks [1]–[4]. Considerable attention and effort have been paid to the challenging issue of how to optimize the required suspension performances, namely, ride comfort, road handling, and suspension deflection. Many vehicle models and controller design methods have been reported in [5]–[7]. On the other hand, many active suspension control approaches have been presented to handle the tradeoff by the utilization of various control techniques, such as fuzzy logic and neural-network control [8], gain scheduling control [9], linear optimal control [10], adaptive control [11], and  $H_\infty$  control [2], [12], [13], and their combined methods.

It should be noted, however, that all the aforementioned suspension control results are under a full reliability assumption that all control components of the systems are in ideal working conditions. Because of the growing complexity of automated control systems, various faults are likely to be encountered, especially faults from actuators and sensors [14]–[16]. Over the past few decades, much research has been attempted to resolve the reliable and fault-tolerant control problems for dynamic systems with uncertainty, such as actuator and sensor faults; a great number of theoretic results have also been presented [17]–[20]. For instance, the reliable  $H_\infty$  controller design problem was investigated in the context of linear systems [21], and a controller was designed ensuring the reliability of the resulting control system, i.e., guaranteed asymptotic stability and  $H_\infty$  performance, under the assumption that all control components of sensors and actuators are operational. As a matter of fact, an active suspension system is different from its counterpart of a passive suspension system in that its actuator has the capability of adjusting force to meet the criteria of the vehicle dynamics, such as guaranteeing the stability of the vehicle, securing passenger comfort, and satisfying the suspension performance.

However, when either the actuator or sensor faults occur in an active suspension system, the conventional controllers cannot achieve better performance in comparison with the reliable and fault-tolerant controllers, as discussed in [22]. Therefore, it is challenging to design a reliable controller such that the system stability and performance of the active suspension closed-loop system can be tolerated with sensor or actuator faults. Because of the electrical and electromagnetic characteristics of the actuators and transmission of the measurement information, electrohydraulic actuators are preferably employed to track the desired forces in order to avoid input time delays; it is a common key factor that degrades the control performances and even causes

Manuscript received April 18, 2011; revised August 19, 2011; accepted September 25, 2011. Date of publication October 31, 2011; date of current version April 4, 2012. This work was supported in part by the 973 Project under Grant 2009CB320600, the National Science Foundation for Distinguished Young Scholars of China under Grant 60825303, the Foundation for the Author of National Excellent Doctoral Dissertation of China under Grant 2007B4, the Key Laboratory of Integrated Automation for the Process Industry, Northeastern University, and a joint Ph.D. scholarship from the University of Portsmouth and Protean Electric, Ltd.

H. Li is with the Research Institute of Intelligent Control and Systems, Harbin Institute of Technology, Harbin 150001, China, and also with the Intelligent Systems and Biomedical Robotics Group, School of Creative Technologies, University of Portsmouth, Portsmouth, PO5 4BP, U.K. (e-mail: hongyi.li@port.ac.uk).

H. Liu is with the Intelligent Systems and Biomedical Robotics Group, School of Creative Technologies, University of Portsmouth, Portsmouth, PO5 4BP, U.K. (e-mail: honghai.liu@port.ac.uk).

H. Gao is with the Research Institute of Intelligent Control and Systems, Harbin Institute of Technology, Harbin 150001, China (e-mail: hjgao@hit.edu.cn).

P. Shi is with the Department of Computing and Mathematical Sciences, University of Glamorgan, Pontypridd, CF37 1DL, U.K., and also with the School of Engineering and Science, Victoria University, Melbourne, Vic. 8001, Australia (e-mail: pshi@glam.ac.uk).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TFUZZ.2011.2174244

instability in the control systems. Recently, controller design schemes have been presented for linear systems with different types of delays [23]–[25]. There exist two mainstreams of controller design methods that involve actuator delays. One is to design a controller by the usage of the integrated system model including actuator dynamics [26]; the other is to consider the actuator time delay in the controller design process in order to design a controller that can stabilize the system and guarantee the closed-loop performance, in spite of the existence of time delay [27].

An active suspension system has the ability to enhance vehicle dynamics by relaxing external impact, such as road surface, on vehicle travel comfort. In terms of its control design, the uncertainty of vehicle sprung and unsprung masses, such as loading conditions, should be taken into account to meet vehicle travel performance criteria. For instance, the polytopic parameter uncertainties were employed to model the varying vehicle sprung or unsprung masses [2], [28], [29]. The parameter-dependent controllers were proposed for the quarter-car suspension systems with sprung mass variation [29]. The parameter-independent sampled-data  $H_\infty$  controller design strategy was presented to handle both sprung and unsprung mass variations in a case study of a quarter-car suspension system [2]. The state of the art in suspension control design in these scenarios, however, could not provide feasible performance for uncertain active suspension systems with actuator delay and fault. Clearly, there is a need for a new controller design method that has the capability of satisfying the control condition. On the other hand, since fuzzy sets were proposed in [30], the fuzzy logic control has developed into a conspicuous and successful branch of automation and control theory. The Takagi–Sugeno (T–S) fuzzy model has been proved as an effective theoretical method and practical tool for the representation of complex nonlinear systems and applications [31]–[34].

The T–S fuzzy-model-based systems are described as a weighted sum of some simple linear subsystems and, thus, are easily analyzable; the success on control analysis and synthesis problems have also been demonstrated by various techniques [35]–[40]. Recently, research has been conducted to challenge the reliability for the continuous-time T–S fuzzy systems [41]–[44]. However, in the context of vehicle suspension control design, there are few results on reliable fuzzy  $H_\infty$  controller design for T–S fuzzy systems with both actuator delay and fault. On the other hand, fuzzy controller design has been investigated for suspension systems over the past few years, e.g., [45]–[47]. In particular, a T–S model-based fuzzy control design approach was presented for electrohydraulic active suspension systems with input constraints [47]. It is evident, however, that there are few results on fuzzy  $H_\infty$  controller design for uncertain active suspension systems with actuator delay and fault.

This paper is concerned with the problem of reliable fuzzy  $H_\infty$  control for uncertain active suspension systems with actuator delay and fault based on the T–S fuzzy model approach. The vehicle dynamic system is established by the fact that vehicle sprung and unsprung mass variations, the actuator delay, and fault have been taken account into the suspension performances.

The parallel-distributed compensation (PDC) scheme is, then, used to develop the reliable fuzzy  $H_\infty$  performance analysis condition for the proposed T–S fuzzy system; the reliable fuzzy  $H_\infty$  controller is designed to guarantee the system's asymptotic stability and  $H_\infty$  performance, simultaneously satisfying the constraint performances. Furthermore, the linear matrix inequality (LMI)-based condition of reliable fuzzy  $H_\infty$  controller design is derived. Finally, the proposed method is evaluated on a quarter-car suspension model. Simulation results demonstrate that the designed reliable fuzzy  $H_\infty$  controller has the robust capability of guaranteeing better suspension performance with uncertainty of the sprung and unsprung mass variations, the actuator delay, and fault.

The remainder of this paper is organized as follows. The problem to be addressed is formulated in Section II. In Section III, we present the reliable fuzzy  $H_\infty$  controller design results, and in Section IV, we provide the fuzzy  $H_\infty$  controller design scheme. Simulation results are provided to evaluate the proposed method in Section V. Finally, the paper is concluded in Section VI.

*Notation:* The notations used throughout the paper are presented as follows. The superscript “ $T$ ” stands for matrix transposition. “ $\mathbb{R}^n$ ” denotes the  $n$ -dimensional Euclidean space. “ $\|\cdot\|_\infty$ ” denotes the  $H_\infty$  norm for matrices. The notation  $P < 0$  ( $\geq 0$ ) denotes a symmetric and positive-definite (semidefinite) matrix. In symmetric block matrices or complex matrix expressions, an asterisk “ $*$ ” is employed to represent a term that is readily induced by symmetry, and  $\text{diag}\{\dots\}$  stands for a block-diagonal matrix. “ $\text{sym}(A)$ ” denotes  $A + A^T$  for simplicity. Matrices, if the dimensions are not explicitly stated, are assumed to be compatible for algebraic operations. The space of square-integrable vector functions over  $[0, \infty)$  is denoted by  $L_2[0, \infty)$ , and for  $w = \{w(t)\} \in L_2[0, \infty)$ , its norm is denoted by  $\|w\|_2 = \sqrt{\int_{t=0}^{\infty} |w(t)|^2 dt}$ .

## II. PROBLEM FORMULATION

There is a substantially growing interest in investigating the modeling and active control design for active suspension systems in the past three decades. It is due to the fact that these systems play an important role in ensuring the suspension performance, such as ride comfort, road holding, and suspension deflection. A quarter-vehicle model has been used widely in the literature to design an active suspension controller, as shown in Fig. 1, where  $m_s$  denotes the sprung mass, which represents the car chassis;  $m_u$  is the unsprung mass, which represents the mass of the wheel assembly;  $u(t)$  stands for the active input of the suspension system;  $z_s$  and  $z_u$  denote the displacements of the sprung and unsprung masses, respectively;  $z_r$  denotes the road displacement input;  $c_s$  and  $k_s$  are damping and stiffness of the suspension system, respectively;  $k_t$  and  $c_t$  stand for compressibility and damping of the pneumatic tire, respectively. Then, the dynamic equation of the suspension model is established as follows:

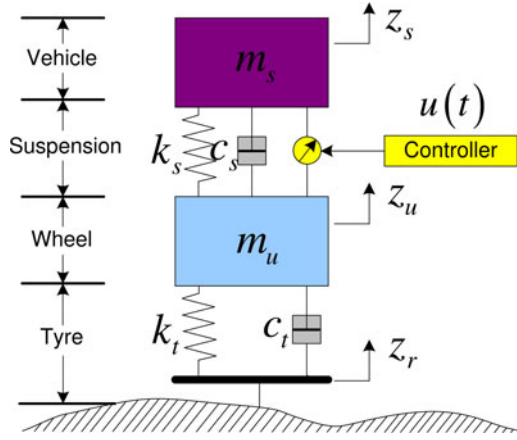


Fig. 1. Quarter-car model.

$$\begin{aligned}
 m_u \ddot{z}_u(t) + c_s [\dot{z}_u(t) - \dot{z}_s(t)] + k_s [z_u(t) - z_s(t)] \\
 + k_t [z_u(t) - z_r(t)] + c_t [\dot{z}_u(t) - \dot{z}_r(t)] \\
 = -u(t) \\
 m_s \ddot{z}_s(t) + c_s [\dot{z}_s(t) - \dot{z}_u(t)] + k_s [z_s(t) - z_u(t)] \\
 = u(t).
 \end{aligned} \quad (1)$$

Let  $x_1(t) = z_s(t) - z_u(t)$  denote the suspension deflection,  $x_2(t) = z_u(t) - z_r(t)$  the tire deflection,  $x_3(t) = \dot{z}_s(t)$  the sprung mass speed,  $x_4(t) = \dot{z}_u(t)$  the unsprung mass speed, and  $w(t) = \dot{z}_r(t)$  the disturbance input. Equation (1) can be rewritten as

$$\dot{x}(t) = A(t)x(t) + B_1(t)w(t) + B(t)u(t) \quad (2)$$

where

$$A(t) = \begin{bmatrix} 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ -\frac{k_s}{m_s} & 0 & -\frac{c_s}{m_s} & \frac{c_s}{m_s} \\ \frac{k_s}{m_u} & -\frac{k_t}{m_u} & \frac{c_s}{m_u} & -\frac{c_s + c_t}{m_u} \end{bmatrix}$$

$$B(t) = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{m_s} \\ -\frac{1}{m_u} \end{bmatrix}, \quad B_1(t) = \begin{bmatrix} 0 \\ -1 \\ 0 \\ \frac{c_t}{m_u} \end{bmatrix}$$

$$x(t) = [x_1(t) \quad x_2(t) \quad x_3(t) \quad x_4(t)]^T.$$

For the control design problems of suspension systems, their performances, such as ride comfort, suspension deflection, and road holding, are the fundamentals that are taken into account. It is widely accepted that ride comfort can be generally quantified by the body acceleration in the vertical direction in the context of a quarter-vehicle model; hence, it is practical to choose body acceleration  $\ddot{z}_s(t)$  as the first control output. It indicates that one of the objectives is to minimize the vertical acceleration  $\ddot{z}_s(t)$  to secure vehicle travel comfort.

Recall the  $H_\infty$  control method; the value of the  $H_\infty$  norm is defined as an upper bound of the root-mean-square (RMS) gain, and the main objective is to minimize the  $H_\infty$  norm of the transfer function from the disturbance  $w(t)$  to the control output, i.e.,  $z_1(t) = \ddot{z}_s(t)$ , with an emphasis on the improvement of ride comfort. Meanwhile, the following required performances have to be taken into account as well.

1) The suspension deflection cannot exceed a maximum value that is constrained by mechanical structure

$$|z_s(t) - z_u(t)| \leq z_{\max} \quad (3)$$

where  $z_{\max}$  is the maximum suspension deflection.

2) The dynamic tire load has to be less than the static tire load in order to ensure a firm uninterrupted contact of the wheels on the road

$$k_t(z_u(t) - z_r(t)) < (m_s + m_u)g. \quad (4)$$

Based on the aforementioned criteria, the body acceleration  $\ddot{z}_s(t)$  is chosen as the performance control output, and the suspension stroke  $z_s(t) - z_u(t)$  and relative dynamic tire load  $k_t(z_u(t) - z_r(t)) / (m_s + m_u)g$  are chosen as the second control output  $z_2(t)$ . Therefore, the following system is derived to present the active vehicle suspension system:

$$\begin{aligned}
 \dot{x}(t) &= A(t)x(t) + B_1(t)w(t) + B(t)u(t) \\
 z_1(t) &= C_1(t)x(t) + D_1(t)u(t) \\
 z_2(t) &= C_2(t)x(t)
 \end{aligned} \quad (5)$$

where  $A(t)$ ,  $B_1(t)$ , and  $B(t)$  are defined in (2), and

$$C_1(t) = \begin{bmatrix} -\frac{k_s}{m_s} & 0 & -\frac{c_s}{m_s} & \frac{c_s}{m_s} \end{bmatrix}, \quad D_1(t) = \frac{1}{m_s}$$

$$C_2(t) = \begin{bmatrix} \frac{1}{z_{\max}} & 0 & 0 & 0 \\ 0 & \frac{k_t}{(m_s + m_u)g} & 0 & 0 \end{bmatrix}. \quad (6)$$

Note that the suspension system in (5) is a model with uncertainty in that the sprung mass  $m_s$  and the unsprung mass  $m_u$  vary in the given ranges, where  $m_s$  and  $m_u$  denote  $m_s(t)$  and  $m_u(t)$ , respectively. In the meantime, the actuator delay and fault should be taken into account, since the suspension performance could be affected by these factors. It leads to the system as

$$\begin{aligned}
 \dot{x}(t) &= A(t)x(t) + B_1(t)w(t) + B(t)u_f(t - d(t)) \\
 z_1(t) &= C_1(t)x(t) + D_1(t)u_f(t - d(t)) \\
 z_2(t) &= C_2(t)x(t) \\
 x(t) &= \phi(t), \quad t \in [-\bar{d}, 0]
 \end{aligned} \quad (7)$$

where  $\phi(t)$  is a vector-valued initial continuous function that is defined on  $t \in [-\bar{d}, 0]$ .  $d(t)$  denotes the time-varying delay satisfying

$$0 \leq d(t) \leq \bar{d}, \quad d(t) \leq \mu. \quad (8)$$

Considering the fault channel from the controller to the actuator

$$u^f(t) = m_a u(t) \quad (9)$$

where  $m_a$  represents the possible fault of the corresponding actuator  $u^f(t)$ .  $\tilde{m}_a \leq m_a \leq \hat{m}_a$ , where  $\tilde{m}_a$  and  $\hat{m}_a$  are the constant scalars and are used to constrain lower and upper bounds of the actuator faults. The following three cases are considered corresponding to three different actuator conditions.

1)  $\tilde{m}_a = \hat{m}_a = 0$ ; then,  $m_a = 0$ , which implies that the corresponding actuator  $u^f(t)$  has completely failed.

2)  $\tilde{m}_a = \hat{m}_a = 1$ ; thus, we obtain  $m_a = 1$ , which represents the case of no fault in the actuator  $u^f(t)$ .

3)  $0 < \tilde{m}_a < \hat{m}_a < 1$ , which means that there exists partial fault in the corresponding actuator  $u^f(t)$ .

The sprung mass  $m_s(t)$  and the unsprung mass  $m_u(t)$  are uncertainties, which vary in a given range, i.e.,  $m_s(t) \in [m_{s\min}, m_{s\max}]$  and  $m_u(t) \in [m_{u\min}, m_{u\max}]$ . It is to say that the uncertainty scenarios of the mass  $m_s(t)$  is bounded by its minimum value  $m_{s\min}$  and its maximum value  $m_{s\max}$ . In addition, the mass  $m_u(t)$  is bounded by its minimum value  $m_{u\min}$  and its maximum value  $m_{u\max}$ . Next, we obtain the values of  $1/m_s(t)$  and  $1/m_u(t)$  from  $m_s(t) \in [m_{s\min}, m_{s\max}]$  and  $m_u(t) \in [m_{u\min}, m_{u\max}]$ . Then, we have

$$\begin{aligned} \max \frac{1}{m_s(t)} &= \frac{1}{m_{s\min}} =: \hat{m}_s, & \min \frac{1}{m_s(t)} &= \frac{1}{m_{s\max}} =: \tilde{m}_s \\ \max \frac{1}{m_u(t)} &= \frac{1}{m_{u\min}} =: \hat{m}_u, & \min \frac{1}{m_u(t)} &= \frac{1}{m_{u\max}} =: \tilde{m}_u. \end{aligned}$$

The sector nonlinear method [32] is employed to represent  $1/m_s(t)$  and  $1/m_u(t)$  by

$$\begin{aligned} \frac{1}{m_s(t)} &= M_1(\xi_1(t)) \hat{m}_s + M_2(\xi_1(t)) \tilde{m}_s \\ \frac{1}{m_u(t)} &= N_1(\xi_2(t)) \hat{m}_u + N_2(\xi_2(t)) \tilde{m}_u \end{aligned}$$

where  $\xi_1(t) = 1/m_s(t)$  and  $\xi_2(t) = 1/m_u(t)$  are the premise variables, and

$$M_1(\xi_1(t)) + M_2(\xi_1(t)) = 1$$

$$N_1(\xi_2(t)) + N_2(\xi_2(t)) = 1.$$

The membership functions  $M_1(\xi_1(t))$ ,  $M_2(\xi_1(t))$ ,  $N_1(\xi_2(t))$ , and  $N_2(\xi_2(t))$  can be calculated as

$$\begin{aligned} M_1(\xi_1(t)) &= \frac{\frac{1}{m_s(t)} - \tilde{m}_s}{\hat{m}_s - \tilde{m}_s}, & M_2(\xi_1(t)) &= \frac{\hat{m}_s - \frac{1}{m_s(t)}}{\hat{m}_s - \tilde{m}_s} \\ N_1(\xi_2(t)) &= \frac{\frac{1}{m_u(t)} - \tilde{m}_u}{\hat{m}_u - \tilde{m}_u}, & N_2(\xi_2(t)) &= \frac{\hat{m}_u - \frac{1}{m_u(t)}}{\hat{m}_u - \tilde{m}_u}. \end{aligned}$$

The member functions are labeled as ‘‘Heavy,’’ ‘‘Light,’’ ‘‘Heavy,’’ and ‘‘Light,’’ as shown in Fig. 2. Then, the system with uncertainty in (7) is represented by the following fuzzy models.

**Model Rule 1:** IF  $\xi_1(t)$  is Heavy and  $\xi_2(t)$  is Heavy,  
THEN

$$\begin{aligned} \dot{x}(t) &= A_1 x(t) + B_1 u_f(t - d(t)) + B_{11} w(t) \\ z_1(t) &= C_{11} x(t) + D_{11} u_f(t - d(t)) \\ z_2(t) &= C_{21} x(t). \end{aligned}$$

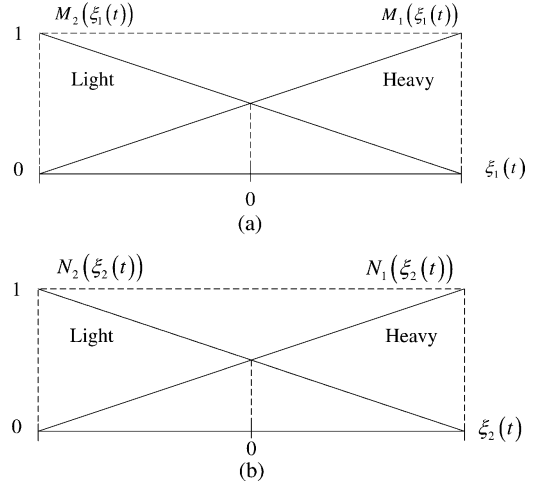


Fig. 2. (a) Membership functions  $M_1(\xi_1(t))$  and  $M_2(\xi_1(t))$ . (b) Membership functions  $N_1(\xi_2(t))$  and  $N_2(\xi_2(t))$ .

Here, by using  $\hat{m}_s$  and  $\hat{m}_u$  to replace  $\frac{1}{m_s(t)}$  and  $\frac{1}{m_u(t)}$  in matrices  $A(t)$ ,  $B(t)$ ,  $B_1(t)$ ,  $C_1(t)$ ,  $D_1(t)$ , and  $C_2(t)$ , we can obtain matrices  $A_1$ ,  $B_1$ ,  $B_{11}$ ,  $C_{11}$ ,  $D_{11}$ , and  $C_{21}$  respectively.

**Model Rule 2:** IF  $\xi_1(t)$  is Heavy and  $\xi_2(t)$  is Light,  
THEN

$$\begin{aligned} \dot{x}(t) &= A_2 x(t) + B_2 u_f(t - d(t)) + B_{12} w(t) \\ z_1(t) &= C_{12} x(t) + D_{12} u_f(t - d(t)) \\ z_2(t) &= C_{22} x(t). \end{aligned}$$

Here, by using  $\hat{m}_s$  and  $\tilde{m}_u$  to replace  $\frac{1}{m_s(t)}$  and  $\frac{1}{m_u(t)}$  in matrices  $A(t)$ ,  $B(t)$ ,  $B_1(t)$ ,  $C_1(t)$ ,  $D_1(t)$ , and  $C_2(t)$ , we can obtain matrices  $A_2$ ,  $B_2$ ,  $B_{12}$ ,  $C_{12}$ ,  $D_{12}$ , and  $C_{22}$  respectively.

**Model Rule 3:** IF  $\xi_1(t)$  is Light and  $\xi_2(t)$  is Heavy,  
THEN

$$\begin{aligned} \dot{x}(t) &= A_3 x(t) + B_3 u_f(t - d(t)) + B_{13} w(t) \\ z_1(t) &= C_{13} x(t) + D_{13} u_f(t - d(t)) \\ z_2(t) &= C_{23} x(t). \end{aligned}$$

Here, by using  $\tilde{m}_s$  and  $\hat{m}_u$  to replace  $\frac{1}{m_s(t)}$  and  $\frac{1}{m_u(t)}$  in matrices  $A(t)$ ,  $B(t)$ ,  $B_1(t)$ ,  $C_1(t)$ ,  $D_1(t)$ , and  $C_2(t)$ , we can obtain matrices  $A_3$ ,  $B_3$ ,  $B_{13}$ ,  $C_{13}$ ,  $D_{13}$ , and  $C_{23}$  respectively.

**Model Rule 4:** IF  $\xi_1(t)$  is Light and  $\xi_2(t)$  is Light,  
THEN

$$\begin{aligned} \dot{x}(t) &= A_4 x(t) + B_4 u_f(t - d(t)) + B_{14} w(t) \\ z_1(t) &= C_{14} x(t) + D_{14} u_f(t - d(t)) \\ z_2(t) &= C_{24} x(t). \end{aligned}$$

Here, by using  $\tilde{m}_s$  and  $\tilde{m}_u$  to replace  $\frac{1}{m_s(t)}$  and  $\frac{1}{m_u(t)}$  in matrices  $A(t)$ ,  $B(t)$ ,  $B_1(t)$ ,  $C_1(t)$ ,  $D_1(t)$  and  $C_2(t)$ , we can obtain matrices  $A_4$ ,  $B_4$ ,  $B_{14}$ ,  $C_{14}$ ,  $D_{14}$ , and  $C_{24}$  respectively.

Fuzzy blending allows us to infer the overall fuzzy model as follows:

$$\begin{aligned}\dot{x}(t) &= \sum_{i=1}^4 h_i(\xi(t)) [A_i x(t) + B_i u_f(t-d(t)) + B_{1i} w(t)] \\ z_1(t) &= \sum_{i=1}^4 h_i(\xi(t)) [C_{1i} x(t) + D_{1i} u_f(t-d(t))] \\ z_2(t) &= \sum_{i=1}^4 h_i(\xi(t)) C_{2i} x(t)\end{aligned}\quad (10)$$

where

$$\begin{aligned}h_1(\xi(t)) &= M_1(\xi_1(t)) \times N_1(\xi_2(t)) \\ h_2(\xi(t)) &= M_1(\xi_1(t)) \times N_2(\xi_2(t)) \\ h_3(\xi(t)) &= M_2(\xi_1(t)) \times N_1(\xi_2(t)) \\ h_4(\xi(t)) &= M_2(\xi_1(t)) \times N_2(\xi_2(t)).\end{aligned}$$

It is apparent that the fuzzy weighting functions  $h_i(\xi(t))$  satisfy  $h_i(\xi(t)) \geq 0$ , and  $\sum_{i=1}^4 h_i(\xi(t)) = 1$ . In order to design a fuzzy reliable controllers, PDC is adapted and the following fuzzy control rules are obtained.

*Control Rule 1:* IF  $\xi_1(t)$  is Heavy and  $\xi_2(t)$  is Heavy, THEN  $u(t) = K_{a1}x(t)$ .

*Control Rule 2:* IF  $\xi_1(t)$  is Heavy and  $\xi_2(t)$  is Light, THEN  $u(t) = K_{a2}x(t)$ .

*Control Rule 3:* IF  $\xi_1(t)$  is Light and  $\xi_2(t)$  is Heavy, THEN  $u(t) = K_{a3}x(t)$ .

*Control Rule 4:* IF  $\xi_1(t)$  is Light and  $\xi_2(t)$  is Light, THEN  $u(t) = K_{a4}x(t)$ .

Hence, the overall fuzzy control law is represented as

$$u(t) = \sum_{j=1}^4 h_j(\xi(t)) K_{aj} x(t) \quad (11)$$

where  $K_{aj}$  ( $j = 1, 2, 3, 4$ ) are the local control gains, and  $u(t-d(t)) = \sum_{j=1}^4 h_j(\xi(t-d(t))) K_{aj} x(t-d(t))$ . Therefore, in this paper, we assume that  $h_j(\xi(t-d(t)))$  is well defined for  $t \in [-\bar{d}, 0]$ , and  $h_j(\xi(t-d(t))) \geq 0$  ( $j = 1, 2, 3, 4$ ),  $\sum_{j=1}^4 h_j(\xi(t-d(t))) = 1$ . For simplicity, the following notations will be used:

$$h_i =: h_i(\xi(t)), \quad h_j^d =: h_j(\xi(t-d(t))).$$

Applying the fuzzy controller (11) to system (10) yields the closed-loop system

$$\begin{aligned}\dot{x}(t) &= \sum_{i=1}^4 \sum_{j=1}^4 h_i h_j^d [A_i x(t) + B_i m_a(t) K_{aj} x(t-d(t)) \\ &\quad + B_{1i} w(t)] \\ z_1(t) &= \sum_{i=1}^4 \sum_{j=1}^4 h_i h_j^d [C_{1i} x(t) + D_{1i} m_a K_{aj} x(t-d(t))] \\ z_2(t) &= \sum_{i=1}^4 h_i C_{2i} x(t).\end{aligned}\quad (12)$$

The T-S fuzzy system in (12) is established based on the practically measurable sprung  $m_s(t)$  and unsprung  $m_u(t)$ . The sector nonlinearity method [32] is employed to analyze the variation of the sprung  $m_s(t)$  and unsprung  $m_u(t)$  and present the T-S fuzzy system in (12).

Without loss of generality, it is assumed that  $w \in L_2[0, \infty)$ , and  $\|w\|_2^2 \leq w_{\max} < \infty$ . The objective in this section is to design the feedback gain matrices  $K_{aj}$  ( $j = 1, 2, 3, 4$ ) such that the following requirements are satisfied.

1) The closed-loop system is asymptotically stable.

2) Under the zero initial condition, the closed-loop system guarantees that  $\|z_1\|_2 < \gamma \|w\|_2$  for all nonzero  $w \in L_2[0, \infty)$ , where  $\gamma > 0$  is a prescribed scalar.

3) The following control output constraints are guaranteed:

$$|\{z_2(t)\}_q| \leq 1, \quad q = 1, 2. \quad (13)$$

### III. RELIABLE FUZZY CONTROLLER DESIGN

In this section, the reliable fuzzy  $H_\infty$  state-feedback controller is derived for the active suspension system with actuator delay and fault. It ensures that the closed-loop system in (12) is asymptotically stable, and it also guarantees a prescribed gain from disturbance  $w(t)$  to performance output  $z_1(t)$ , under the condition that the suspension stroke and tire deflection constraints are satisfied. First, the following lemma is presented.

*Lemma 1:* [21] For a time-varying diagonal matrix, i.e.,  $\Phi(t) = \text{diag}\{\sigma_1(t), \sigma_2(t), \dots, \sigma_p(t)\}$ , and two matrices  $R$  and  $S$  with appropriate dimensions, if  $|\Phi(t)| \leq V$ , where  $V < 0$  is a known diagonal matrix, then for any scalar  $\varepsilon > 0$ , it is true that

$$R\Phi S + S^T \Phi^T R^T \leq \varepsilon R V R^T + \varepsilon^{-1} S^T V S.$$

Next, the following scalars are introduced which will be used in the later development in this paper.  $M_{a0} = (\hat{m}_a + \hat{n}_a)/2$ ,  $L_a = [m_a - M_{a0}]/m_{a0}$ , and  $J_a = (\hat{m}_a - \hat{n}_a)/(\hat{m}_a + \hat{n}_a)$ . Thus, one has  $m_a = M_{a0}(I + L_a)$  and  $L_a^T L_a \leq J_a^T J_a \leq I$ . Then, it leads to the following theorem.

*Theorem 1:* Consider the closed-loop system in (12). For given scalars  $\bar{d} > 0$ ,  $\mu$ , and matrices  $K_{aj}$ , if there exist matrices  $P < 0$ ,  $Q < 0$ ,  $S < 0$ ,  $R < 0$ ,  $N_j$ , and  $M_j$  with appropriate dimensions and positive scalars  $\varepsilon_{1ij} < 0$  and  $\varepsilon_{2ij} < 0$  ( $i, j = 1, 2, 3, 4$ ), such that the following LMIs hold for  $q = 1, 2$ :

$$\begin{bmatrix} \Phi_{11}^{ij} & \sqrt{\bar{d}}M & \Phi_{13}^{ij} & \Phi_{14}^{ij} & \Phi_{15}^{ij} & \Phi_{16}^{1ij} \\ 0 & -R & 0 & 0 & 0 & 0 \\ 0 & 0 & -I & 0 & D_{1i} & 0 \\ 0 & 0 & 0 & -R & \sqrt{\bar{d}}RB_i & 0 \\ 0 & 0 & 0 & 0 & -\varepsilon_{1ij}J_a^{-1} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\varepsilon_{1ij}J_a^{-1} \end{bmatrix} < 0 \quad (14)$$

$$\begin{bmatrix} \Phi_{11}^{ij} & \sqrt{\bar{d}}N & \Phi_{13}^{ij} & \Phi_{14}^{ij} & \Phi_{15}^{ij} & \Phi_{16}^{2ij} \\ 0 & -R & 0 & 0 & 0 & 0 \\ 0 & 0 & -I & 0 & D_{1i} & 0 \\ 0 & 0 & 0 & -R & \sqrt{\bar{d}}RB_i & 0 \\ 0 & 0 & 0 & 0 & -\varepsilon_{2ij}J_a^{-1} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\varepsilon_{2ij}J_a^{-1} \end{bmatrix} < 0 \quad (15)$$

$$\begin{bmatrix} -P & \sqrt{\rho}\{C_{2i}\}_q^T \\ * & -I \end{bmatrix} < 0 \quad (16)$$

where

$$\begin{aligned} \Phi_{11}^{ij} &= \Xi_{11}^{ij} + \text{sym}(\Xi_2), \quad \Xi_{11}^{ij} = \begin{bmatrix} \Theta_{11}^{ij} & \Theta_{12}^{ij} \\ * & -\gamma^2 I \end{bmatrix} \\ \Theta_{11}^{ij} &= \begin{bmatrix} \text{sym}(PA_i) + Q + S & PB_i M_{a0} K_{aj} & 0 \\ * & -(1-\mu)S & 0 \\ * & * & -Q \end{bmatrix} \\ \Theta_{12}^{ij} &= \begin{bmatrix} PB_{1i} \\ 0 \\ 0 \end{bmatrix}, \quad \Xi_2 = \begin{bmatrix} M & N-M & -N & 0 \end{bmatrix} \\ \Phi_{13}^{ij} &= [C_{1i} \quad D_{1i} M_{a0} K_{aj} \quad 0 \quad 0]^T \\ \Phi_{14}^{ij} &= [\sqrt{d} R A_i \quad \sqrt{d} R B_i M_{a0} K_{aj} \quad 0 \quad \sqrt{d} R B_{1i}]^T \\ \Phi_{15}^{ij} &= [B_i^T P \quad 0 \quad 0 \quad 0]^T \\ \Phi_{16}^{1ij} &= [0 \quad \varepsilon_{1ij} M_{a0} K_{aj} \quad 0 \quad 0]^T \\ \Phi_{16}^{2ij} &= [0 \quad \varepsilon_{2ij} M_{a0} K_{aj} \quad 0 \quad 0]^T \\ M &= [M_1^T \quad M_2^T \quad M_3^T \quad M_4^T]^T \\ N &= [N_1^T \quad N_2^T \quad N_3^T \quad N_4^T]^T. \end{aligned}$$

Furthermore, 1) the closed-loop system is asymptotically stable, and 2) the performance  $\|T_{z_1 w}\|_\infty < \gamma$  is minimized subject to output constraints (13) with the disturbance energy under the bound, i.e.,  $w_{\max} = (\rho - V(0))/\gamma^2$ , where  $T_{z_1 w}$  denotes the closed-loop transfer function from the road disturbance  $w(t)$  to the control output  $z_1(t)$ .

*Proof:* Consider the Lyapunov–Krasovskii functional as follows:

$$\begin{aligned} V(t) &= x^T(t) P x(t) + \int_{t-\bar{d}}^t x^T(s) Q x(s) ds \\ &\quad + \int_{t-d(t)}^t x^T(s) S x(s) ds \\ &\quad + \int_{-\bar{d}}^0 \int_{t+\alpha}^t \dot{x}^T(s) R \dot{x}(s) ds d\alpha. \end{aligned} \quad (17)$$

The derivative of  $V(t)$  along the solution of system (12) is expressed as

$$\begin{aligned} \dot{V}(t) &\leq 2x^T(t) P \dot{x}(t) + x^T(t) (Q + S) x(t) \\ &\quad - x^T(t-\bar{d}) Q x(t-\bar{d}) + \bar{d} \dot{x}^T(t) S \dot{x}(t) \\ &\quad - (1-\mu) x^T(t-d(t)) Q x(t-d(t)) \\ &\quad - \int_{t-d(t)}^t \dot{x}^T(s) R \dot{x}(s) ds \\ &\quad - \int_{t-\bar{d}}^{t-d(t)} \dot{x}^T(s) R \dot{x}(s) ds. \end{aligned} \quad (18)$$

To develop  $H_\infty$  performance analysis criterion, the system (12) is stable with  $w(t) = 0$ ; then, the  $H_\infty$  performance index is satisfied. For any appropriately dimensioned matrices  $\hat{M}$  and  $\hat{N}$ , the following equalities hold directly according to the Newton–Leibniz formula:

$$\begin{aligned} \eta_1(t) &= 2\xi^T(t) \hat{M} \\ &\quad \times \left( x(t) - x(t-d(t)) - \int_{t-d(t)}^t \dot{x}(s) ds \right) = 0 \\ \eta_2(t) &= 2\xi^T(t) \hat{N} \\ &\quad \times \left( x(t-d(t)) - x(t-\bar{d}) - \int_{t-\bar{d}}^{t-d(t)} \dot{x}(s) ds \right) = 0 \end{aligned}$$

where

$$\begin{aligned} \xi^T(t) &= [x^T(t) \quad x^T(t-d(t)) \quad x^T(t-\bar{d})] \\ \hat{M} &= [M_1^T \quad M_2^T \quad M_3^T]^T, \quad \hat{N} = [N_1^T \quad N_2^T \quad N_3^T]^T. \end{aligned}$$

Adding  $\eta_1(t)$  and  $\eta_2(t)$  to the right-hand side of (18), the following inequalities are obtained:

$$\begin{aligned} \dot{V}(t) &\leq \sum_{i=1}^4 \sum_{j=1}^4 h_i h_j^d \xi^T(t) \left[ \hat{\Xi}_{ij} + d(t) \hat{M} R^{-1} \hat{M}^T \right. \\ &\quad \left. + (\bar{d} - d(t)) \hat{N} R^{-1} \hat{N}^T \right] \xi(t) \\ &\quad - \int_{t-d(t)}^t \left[ \xi^T(t) \hat{M} + \dot{x}^T(s) R \right] R^{-1} \\ &\quad \times \left[ \hat{M}^T \xi(t) + R \dot{x}(s) \right] ds \\ &\quad - \int_{t-\bar{d}}^{t-d(t)} \left[ \xi^T(t) \hat{N} + \dot{x}^T(s) R \right] R^{-1} \\ &\quad \times \left[ \hat{N}^T \xi(t) + R \dot{x}(s) \right] ds \\ &\leq \sum_{i=1}^4 \sum_{j=1}^4 h_i h_j^d \xi^T(t) \left[ \hat{\Xi}_{ij} + d(t) \hat{M} R^{-1} \hat{M}^T \right. \\ &\quad \left. + (\bar{d} - d(t)) \hat{N} R^{-1} \hat{N}^T \right] \xi(t) \\ &= \sum_{i=1}^4 \sum_{j=1}^4 h_i h_j^d \xi^T(t) \left[ \frac{d(t)}{\bar{d}} \left( \hat{\Xi}_{ij} + \bar{d} \hat{M} R^{-1} \hat{M}^T \right) \right. \\ &\quad \left. + \frac{\bar{d} - d(t)}{\bar{d}} \left( \hat{\Xi}_{ij} + \bar{d} \hat{N} R^{-1} \hat{N}^T \right) \right] \xi(t). \end{aligned}$$

Here

$$\hat{\Xi}_{ij} = \hat{\Theta}_{11}^{ij} + \text{sym}(\hat{\Pi}_2) + \Upsilon \bar{d} R \Upsilon^T$$

where

$$\hat{\Pi}_2 = [\hat{M} \quad \hat{N} - \hat{M} \quad -\hat{N}], \quad \Upsilon = [A_i \quad B_i m_a K_{aj} \quad 0]^T$$

and the matrix  $\hat{\Theta}_{11}^{ij}$  is the matrix  $\Theta_{11}^{ij}$ , where the term  $PB_iM_{a0}K_{aj}$  is replaced by  $PB_im_aK_{aj}$ . It is found that

$$\begin{aligned}\tilde{\Xi}_{ij}^1 &= \begin{bmatrix} \hat{\Theta}_{11}^{ij} + \text{sym}(\hat{\Pi}_2) & \sqrt{d}\hat{M} & \sqrt{d}\Upsilon R \\ * & -R & 0 \\ * & * & -R \end{bmatrix} \\ &\leq \begin{bmatrix} \Theta_{11}^{ij} & \sqrt{d}\hat{M} & \tilde{\Phi}_{14}^{ij} \\ * & -R & 0 \\ * & * & -R \end{bmatrix} + \varepsilon_{1ij}^{-1}\Lambda^T J_a \Lambda + \varepsilon_{1ij}\Delta J_a \Delta^T \\ \tilde{\Xi}_{ij}^2 &= \begin{bmatrix} \hat{\Theta}_{11}^{ij} + \text{sym}(\hat{\Pi}_2) & \sqrt{d}\hat{N} & \sqrt{d}\Upsilon R \\ * & -R & 0 \\ * & * & -R \end{bmatrix} \\ &\leq \begin{bmatrix} \Theta_{11}^{ij} & \sqrt{d}\hat{N} & \tilde{\Phi}_{14}^{ij} \\ * & -R & 0 \\ * & * & -R \end{bmatrix} + \varepsilon_{2ij}^{-1}\Lambda^T J_a \Lambda + \varepsilon_{2ij}\Delta J_a \Delta^T\end{aligned}$$

and

$$\begin{aligned}\tilde{\Phi}_{14}^{ij} &= [\sqrt{d}RA_i \quad \sqrt{d}RB_iM_{a0}K_{aj} \quad 0]^T \\ \Lambda &= [B_i^T P \quad 0 \quad 0 \quad \sqrt{d}B_i^T R \quad 0] \\ \Delta^T &= [0 \quad M_{a0}K_{aj} \quad 0 \quad 0 \quad 0].\end{aligned}$$

From (14) and (15) and according to the Schur complement,  $\tilde{\Xi}_{ij}^1 < 0$  and  $\tilde{\Xi}_{ij}^2 < 0$  are obtained; this is to say that

$$\hat{\Xi}_{ij} + \bar{d}\hat{M}R^{-1}\hat{M}^T < 0, \quad \hat{\Xi}_{ij} + \bar{d}\hat{N}R^{-1} < 0$$

which leads to  $\dot{V}(t) < 0$ ; then, the system (12) is asymptotically stable for the delay  $d(t)$ , satisfying (8). Next, the  $H_\infty$  performance of the system (12) is established under zero initial conditions. First, the Lyapunov functional is defined as shown in (17). It is not difficult to achieve

$$\begin{aligned}\dot{V}(t) + z_1^T(t)z_1(t) - \gamma^2 w^T(t)w(t) &\leq \sum_{i=1}^4 \sum_{j=1}^4 h_i h_j^d \bar{\xi}^T(t) [\tilde{\Xi}_{ij} + d(t)MR^{-1}M^T \\ &\quad + (\bar{d} - d(t))NR^{-1}N^T] \bar{\xi}(t) \\ &= \sum_{i=1}^4 \sum_{j=1}^4 h_i h_j^d \bar{\xi}^T(t) \left[ \frac{d(t)}{\bar{d}} (\tilde{\Xi}_{ij} + \bar{d}MR^{-1}M^T) \right. \\ &\quad \left. + \frac{\bar{d} - d(t)}{\bar{d}} (\tilde{\Xi}_{ij} + \bar{d}NR^{-1}N^T) \right] \bar{\xi}(t)\end{aligned}$$

where

$$\begin{aligned}\tilde{\Xi}_{ij} &= \tilde{\Phi}_{11}^{ij} + \tilde{\Phi}_{13}^{ij}\tilde{\Phi}_{13}^{ijT} + \tilde{\Phi}_{14}^{ij}\tilde{\Phi}_{14}^{ijT} \\ \bar{\xi}^T(t) &= [\xi^T(t) \quad w^T(t)]\end{aligned}$$

and  $\tilde{\Phi}_{11}^{ij}$ ,  $\tilde{\Phi}_{13}^{ij}$ , and  $\tilde{\Phi}_{14}^{ij}$  are the matrices  $\Phi_{11}^{ij}$ ,  $\Phi_{13}^{ij}$ , and  $\Phi_{14}^{ij}$  in which the terms  $PB_iM_{a0}K_{aj}$ ,  $K_{aj}^T M_{a0}D_{1i}^T$ , and  $\sqrt{d}K_{aj}^T M_{a0}B_i^T R$  are replaced by the terms  $PB_im_aK_{aj}$ ,  $K_{aj}^T m_a D_{1i}^T$ , and  $K_{aj}^T m_a B_i^T$ , respectively. According to the Schur complement and the aforementioned method, we develop

$$\dot{V}(t) + z_1^T(t)z_1(t) - \gamma^2 w^T(t)w(t) < 0 \quad (19)$$

for all nonzero  $w \in L_2[0, \infty)$ . Under zero initial conditions, we have  $V(0) = 0$  and  $V(\infty) \geq 0$ . Integrating both sides of (19) yields  $\|z_1\|_2 < \gamma \|w\|_2$  for all nonzero  $w \in L_2[0, \infty)$ , and the  $H_\infty$  performance is established.

In what follows, we will show that the hard constraints in (13) are guaranteed. Inequality (19) guarantees  $\dot{V}(t) - \gamma^2 w^T(t)w(t) < 0$ . By the integration of both sides of the aforementioned inequality from zero to any  $t < 0$ , we obtain

$$V(t) - V(0) < \gamma^2 \int_0^t w^T(s)w(s) ds < \gamma^2 \|w\|_2^2. \quad (20)$$

From the definition of the Lyapunov functional in (17), we obtain that  $x^T(t)Px(t) < \rho$  with  $\rho = \gamma^2 w_{\max} + V(0)$ . Similar to [2], the following inequality holds:

$$\begin{aligned}\max_{t>0} |z_2(t)|_q^2 &\leq \max_{t>0} \left\| \sum_{i=1}^4 h_i x^T(t) \{C_{2i}\}_q^T \{C_{2i}\}_q x(t) \right\|_2 \\ &= \max_{t>0} \left\| \sum_{i=1}^4 h_i x^T(t) P^{-\frac{1}{2}} P^{-\frac{1}{2}} \{C_{2i}\}_q^T \{C_{2i}\}_q P^{-\frac{1}{2}} P^{\frac{1}{2}} x(t) \right\|_2 \\ &< \rho \cdot \theta_{\max} \left( \sum_{i=1}^4 h_i P^{-\frac{1}{2}} \{C_{2i}\}_q^T \{C_{2i}\}_q P^{-\frac{1}{2}} \right), \quad q = 1, 2\end{aligned}$$

where  $\theta_{\max}(\cdot)$  represents the maximal eigenvalue. The aforementioned inequality leads to the fact that the constraints in (13) are guaranteed, if

$$\rho \cdot \sum_{i=1}^4 h_i P^{-\frac{1}{2}} \{C_{2i}\}_q^T \{C_{2i}\}_q P^{-\frac{1}{2}} < I \quad (21)$$

which means

$$\sum_{i=1}^4 h_i \left( \rho \cdot P^{-\frac{1}{2}} \{C_{2i}\}_q^T \{C_{2i}\}_q P^{-\frac{1}{2}} - I \right) < 0$$

which is guaranteed by the feasibility of (16). Hence, the proof is complete. ■

*Remark 1:* In this paper, the free-weight matrix method [48] has been utilized to propose the delay-dependent  $H_\infty$  performance analysis condition for the time-varying actuator delay  $d(t)$ . How to develop the less-conservative condition is still a challenging research topic. The interval time-varying delay [49] and present less-conservative results have been targeted in our future work.

In what follows, the reliable fuzzy  $H_\infty$  controller existence condition is presented for the active suspension system in (12), it is based on reliable fuzzy  $H_\infty$  performance analysis criterion in Theorem 1.

*Theorem 2:* Consider the closed-loop system in (12). For given scalars  $\bar{d} < 0$  and  $\mu$ , if there exist matrices  $\bar{P} < 0$ ,  $\bar{Q} < 0$ ,  $\bar{S} < 0$ ,  $\bar{R} < 0$ ,  $Y_{aj}$ ,  $\bar{N}_j$ , and  $\bar{M}_j$  with appropriate dimensions and positive scalars  $\bar{\varepsilon}_{1ij} < 0$  and  $\bar{\varepsilon}_{2ij} < 0$  ( $i, j = 1, 2, 3, 4$ ),

such that the following LMIs hold for  $q = 1, 2$ :

$$\begin{bmatrix} \bar{\Phi}_{11}^{ij} & \sqrt{d}\bar{M} & \bar{\Phi}_{13}^{ij} & \bar{\Phi}_{14}^{ij} & \bar{\Phi}_{15}^{ij} & \bar{\Phi}_{16}^{ij} \\ 0 & \bar{R} - 2\bar{P} & 0 & 0 & 0 & 0 \\ 0 & 0 & -I & 0 & D_{1i} & 0 \\ 0 & 0 & 0 & -\bar{R} & \bar{\varepsilon}_{1ij}\sqrt{d}B_i & 0 \\ 0 & 0 & 0 & 0 & -\bar{\varepsilon}_{1ij}J_a^{-1} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\bar{\varepsilon}_{1ij}J_a^{-1} \end{bmatrix} < 0 \quad (22)$$

$$\begin{bmatrix} \bar{\Phi}_{11}^{ij} & \sqrt{d}\bar{N} & \bar{\Phi}_{13}^{ij} & \bar{\Phi}_{14}^{ij} & \bar{\Phi}_{15}^{ij} & \bar{\Phi}_{16}^{ij} \\ 0 & \bar{R} - 2\bar{P} & 0 & 0 & 0 & 0 \\ 0 & 0 & -I & 0 & D_{1i} & 0 \\ 0 & 0 & 0 & -\bar{R} & \bar{\varepsilon}_{2ij}\sqrt{d}B_i & 0 \\ 0 & 0 & 0 & 0 & -\bar{\varepsilon}_{2ij}J_a^{-1} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\bar{\varepsilon}_{2ij}J_a^{-1} \end{bmatrix} < 0 \quad (23)$$

$$\begin{bmatrix} -\bar{P} & \sqrt{\rho}\bar{P}\{C_{2i}\}_q^T \\ * & -I \end{bmatrix} < 0 \quad (24)$$

where

$$\begin{aligned} \bar{\Phi}_{11}^{ij} &= \bar{\Xi}_{11}^{ij} + \text{sym}(\bar{\Xi}_2), \quad \bar{\Xi}_{11}^{ij} = \begin{bmatrix} \bar{\Theta}_{11}^{ij} & \bar{\Theta}_{12}^{ij} \\ * & -\gamma^2 I \end{bmatrix} \\ \bar{\Theta}_{11}^{ij} &= \begin{bmatrix} \text{sym}(A_i \bar{P}) + \bar{Q} + \bar{S} & B_i Y_{aj} & 0 \\ * & -(1-\mu)\bar{S} & 0 \\ * & * & -\bar{Q} \end{bmatrix} \\ \bar{\Theta}_{12}^{ij} &= \begin{bmatrix} B_{1i} \\ 0 \\ 0 \end{bmatrix}, \quad \bar{\Xi}_2 = [\bar{M} \quad \bar{N} - \bar{M} \quad -\bar{N} \quad 0] \\ \bar{\Phi}_{13}^{ij} &= [C_{1i}\bar{P} \quad D_{1i}Y_{aj} \quad 0 \quad 0]^T \\ \bar{\Phi}_{14}^{ij} &= [\sqrt{d}A_i\bar{P} \quad \sqrt{d}B_iY_{aj} \quad 0 \quad \sqrt{d}B_{1i}]^T \\ \bar{\Phi}_{15}^{ij} &= [\bar{\varepsilon}_{1ij}B_i^T \quad 0 \quad 0 \quad 0]^T \\ \bar{\Phi}_{15}^{2ij} &= [\bar{\varepsilon}_{2ij}B_i^T \quad 0 \quad 0 \quad 0]^T \\ \bar{\Phi}_{16}^{ij} &= [0 \quad Y_{aj} \quad 0 \quad 0]^T \\ \bar{M} &= [\bar{M}_1^T \quad \bar{M}_2^T \quad \bar{M}_3^T \quad \bar{M}_4^T]^T \\ \bar{N} &= [\bar{N}_1^T \quad \bar{N}_2^T \quad \bar{N}_3^T \quad \bar{N}_4^T]^T. \end{aligned}$$

Then, a reliable controller in the form of (11) exists, such that 1) the closed-loop system is asymptotically stable, and 2) the performance  $\|T_{z_1 w}\|_\infty < \gamma$  is minimized subject to output constraints (13) with the disturbance energy under the bound:  $w_{\max} = (\rho - V(0))/\gamma^2$ .

Moreover, if inequalities (22)–(24) have a feasible solution, then the control gain  $K_{aj}$  in (11) is given by  $K_{aj} = M_{a0}^{-1}Y_{aj}\bar{P}^{-1}$ .

*Proof:* From  $(\bar{R} - \bar{P})\bar{R}^{-1}(\bar{R} - \bar{P}) \geq 0$ , we have  $-\bar{P}\bar{R}^{-1}\bar{P} \leq \bar{R} - 2\bar{P}$ . After replacing  $\bar{R} - 2\bar{P}$  in (22)–(23) with

$-\bar{P}\bar{R}^{-1}\bar{P}$  and performing corresponding congruence transformation by

$$\text{diag}\{\bar{P}^{-1}, \bar{P}^{-1}, \bar{P}^{-1}, I, \bar{P}^{-1}, I, \bar{R}^{-1}, \bar{\varepsilon}_{1ij}^{-1}I, \bar{\varepsilon}_{1ij}^{-1}I\}$$

and

$$\text{diag}\{\bar{P}^{-1}, \bar{P}^{-1}, \bar{P}^{-1}, I, \bar{P}^{-1}, I, \bar{R}^{-1}, \bar{\varepsilon}_{2ij}^{-1}I, \bar{\varepsilon}_{2ij}^{-1}I\}$$

together with the change of matrix variables that are defined by

$$P = \bar{P}^{-1}, \quad R = \bar{R}^{-1}, \quad Q = \bar{P}^{-1}\bar{Q}\bar{P}^{-1}$$

$$K_{aj} = M_{a0}^{-1}Y_{aj}\bar{P}^{-1}, \quad S = \bar{P}^{-1}\bar{S}\bar{P}^{-1}, \quad \varepsilon_{1ij} = \bar{\varepsilon}_{1ij}^{-1}$$

$$\varepsilon_{2ij} = \bar{\varepsilon}_{2ij}^{-1}, \quad M = \text{diag}\{\bar{P}^{-1}, \bar{P}^{-1}, \bar{P}^{-1}, I\}\bar{M}\bar{P}^{-1}$$

$$N = \text{diag}\{\bar{P}^{-1}, \bar{P}^{-1}, \bar{P}^{-1}, I\}\bar{N}\bar{P}^{-1}$$

it is concluded that conditions (14) and (15) hold. On the other hand, (24) is equivalent to (16), performing a simple congruence transformation with  $\text{diag}\{\bar{P}^{-1}, I\}$ . Therefore, all the conditions in Theorem 1 are satisfied. Hence, the proof is complete. ■

*Remark 2:* In this study, the conservative will be reduced if the matrices  $Q$ ,  $S$ ,  $R$ ,  $M$ , and  $N$  are replaced by  $\sum_{i=1}^4 h_i Q_i$ ,  $\sum_{i=1}^4 h_i S_i$ ,  $\sum_{i=1}^4 h_i R_i$ ,  $\sum_{i=1}^4 h_i M_i = \sum_{i=1}^4 h_i [M_{1i}^T \ M_{2i}^T \ M_{3i}^T \ M_{4i}^T]^T$ , and  $\sum_{i=1}^4 h_i N_i = \sum_{i=1}^4 h_i [N_{1i}^T \ N_{2i}^T \ N_{3i}^T \ N_{4i}^T]^T$ , respectively. However, computation complexon of the existence condition in Theorem 2 of reliable fuzzy  $H_\infty$  controller design will be increased intensively. Thus, the aforementioned proof is employed to handle the tradeoff in this study.

#### IV. FUZZY $H_\infty$ CONTROLLER DESIGN

In the section, fuzzy  $H_\infty$  controller design is presented for active suspension systems with actuator delay based on the T-S fuzzy model method. If there is no actuator fault in the active suspension system, then we obtain

$$\begin{aligned} \dot{x}(t) &= A(t)x(t) + B_1(t)w(t) + B(t)u(t-d(t)) \\ z_1(t) &= C_1(t)x(t) + D_1(t)u(t-d(t)) \\ z_2(t) &= C_2(t)x(t). \end{aligned} \quad (25)$$

Based on the fuzzy modeling presented earlier, the overall fuzzy model is inferred as follows:

$$\begin{aligned} \dot{x}(t) &= \sum_{i=1}^4 h_i(\xi(t)) [A_i x(t) + B_i u(t-d(t)) + B_{1i} w(t)] \\ z_1(t) &= \sum_{i=1}^4 h_i(\xi(t)) [C_{1i} x(t) + D_i u(t-d(t))] \\ z_2(t) &= \sum_{i=1}^4 h_i(\xi(t)) C_{2i} x(t). \end{aligned} \quad (26)$$

In addition, the overall fuzzy control law is represented as

$$u(t) = \sum_{j=1}^4 h_j(\xi(t)) K_{sj} x(t). \quad (27)$$

For the case of the standard controller (27), the closed-loop system is given by

$$\begin{aligned}\dot{x}(t) &= \sum_{i=1}^4 \sum_{j=1}^4 h_i h_j^d [A_i x(t) + B_i K_{sj} x(t-d(t)) + B_{1i} w(t)] \\ z_1(t) &= \sum_{i=1}^4 \sum_{j=1}^4 h_i h_j^d [C_{1i} x(t) + D_{1i} K_{sj} x(t-d(t))] \\ z_2(t) &= \sum_{i=1}^4 h_i C_{2i} x(t).\end{aligned}\quad (28)$$

Employing the similar method that is proposed in the previous section, the following corollary is obtained for the fuzzy  $H_\infty$  performance analysis at the context of the system (28) with actuator delay.

*Corollary 1* Consider the closed-loop system (28). Given scalars  $\bar{d} > 0$ ,  $\mu$ , and matrices  $K_{sj}$ , if there exist matrices  $P > 0$ ,  $Q > 0$ ,  $S > 0$ ,  $R > 0$ ,  $N_j$ , and  $M_j$  ( $j = 1, 2, 3, 4$ ) with appropriate dimensions, such that the following LMIs hold for  $q = 1, 2$ :

$$\begin{bmatrix} \Phi_{11}^{ij} & \sqrt{\bar{d}}M & \Phi_{13}^{ij} & \Phi_{14}^{ij} \\ 0 & -R & 0 & 0 \\ 0 & 0 & -I & 0 \\ 0 & 0 & 0 & -R \end{bmatrix} < 0 \quad (29)$$

$$\begin{bmatrix} \Phi_{11}^{ij} & \sqrt{\bar{d}}N & \Phi_{13}^{ij} & \Phi_{14}^{ij} \\ 0 & -R & 0 & 0 \\ 0 & 0 & -I & 0 \\ 0 & 0 & 0 & -R \end{bmatrix} < 0 \quad (30)$$

$$\begin{bmatrix} -P & \sqrt{\rho}\{C_{2i}\}_q^T \\ * & -I \end{bmatrix} < 0 \quad (31)$$

where

$$\begin{aligned}\Phi_{11}^{ij} &= \Xi_{11}^{ij} + \text{sym}(\Xi_2), \quad \Xi_{11}^{ij} = \begin{bmatrix} \Theta_{11}^{ij} & \Theta_{12}^{ij} \\ * & -\gamma^2 I \end{bmatrix} \\ \Phi_{13}^{ij} &= [C_{1i} \quad D_{1i} K_{sj} \quad 0 \quad 0]^T \\ \Theta_{11}^{ij} &= \begin{bmatrix} \text{sym}(PA_i) + Q + S & PB_i K_{sj} & 0 \\ * & -(1-\mu)S & 0 \\ * & * & -Q \end{bmatrix} \\ \Phi_{14}^{ij} &= [\sqrt{\bar{d}}RA_i \quad \sqrt{\bar{d}}RB_i K_{sj} \quad 0 \quad \sqrt{\bar{d}}RB_{1i}]^T.\end{aligned}$$

Taking into account the matrices  $\Xi_2$ ,  $\Theta_{12}^{ij}$ ,  $M$ , and  $N$  in Theorem 1, we obtain that 1) the closed-loop system is asymptotically stable, and 2) the performance  $\|T_{z_1 w}\|_\infty < \gamma$  is minimized subject to output constraints (13).

Similarly, the fuzzy  $H_\infty$  controller design condition is derived from Theorem 2.

*Corollary 2:* Consider the closed-loop system (28). Given scalars  $\bar{d} > 0$  and  $\mu$ , the closed-loop system (12) is asymptotically stable with an  $H_\infty$  disturbance attenuation level  $\gamma$ , if there exist matrices  $\bar{P} > 0$ ,  $\bar{Q} > 0$ ,  $\bar{S} > 0$ ,  $\bar{R} > 0$ ,  $Y_{sj}$ ,  $\bar{N}_j$ , and

$\bar{M}_j$  ( $j = 1, 2, 3, 4$ ) with appropriate dimensions, such that the following LMIs hold for  $q = 1, 2$ :

$$\begin{bmatrix} \Phi_{11}^{ij} & \Phi_{12}^{ij} & \Phi_{13}^{ij} & \Phi_{14}^{ij} \\ 0 & \bar{R} - 2\bar{P} & 0 & 0 \\ 0 & 0 & -I & 0 \\ 0 & 0 & 0 & -\bar{R} \end{bmatrix} < 0 \quad (32)$$

$$\begin{bmatrix} \Phi_{11}^{ij} & \Phi_{12}^{ij} & \Phi_{13}^{ij} & \Phi_{14}^{ij} \\ 0 & \bar{R} - 2\bar{P} & 0 & 0 \\ 0 & 0 & -I & 0 \\ 0 & 0 & 0 & -\bar{R} \end{bmatrix} < 0 \quad (33)$$

$$\begin{bmatrix} -\bar{P} & \sqrt{\rho}\bar{P}\{C_{2i}\}_q^T \\ * & -I \end{bmatrix} < 0 \quad (34)$$

where

$$\begin{aligned}\Phi_{11}^{ij} &= \Xi_{11}^{ij} + \text{sym}(\Xi_2), \quad \Xi_{11}^{ij} = \begin{bmatrix} \Xi_{11}^{ij} & \Theta_{12}^{ij} \\ * & -\gamma^2 I \end{bmatrix} \\ \Phi_{13}^{ij} &= [C_{1i}\bar{P} \quad D_{1i}Y_{sj} \quad 0 \quad 0]^T \\ \Theta_{11}^{ij} &= \begin{bmatrix} \text{sym}(A_i\bar{P}) + \bar{Q} + \bar{S} & B_iY_{sj} & 0 \\ * & -(1-\mu)\bar{S} & 0 \\ * & * & -\bar{Q} \end{bmatrix} \\ \Phi_{14}^{ij} &= [\sqrt{\bar{d}}A_i\bar{P} \quad \sqrt{\bar{d}}B_iY_{sj} \quad 0 \quad \sqrt{\bar{d}}B_{1i}]^T\end{aligned}$$

where  $\Xi_2$ ,  $\Theta_{12}^{ij}$ ,  $\bar{M}$ , and  $\bar{N}$  are defined in Theorem 2. Then, a standard controller in the form of (27) exists, such that 1) the closed-loop system is asymptotically stable, and 2) the performance  $\|T_{z_1 w}\|_\infty < \gamma$  is minimized subject to output constraint (13).

Moreover, if inequalities (32)–(34) have a feasible solution, then the control gain  $K_{sj}$  in (27) is given by  $K_{sj} = Y_{sj}\bar{P}^{-1}$ .

*Remark 3:* When the derivative of  $d(t)$  is unknown, and the delay  $d(t)$  satisfies  $0 < d(t) \leq \bar{d}$ , by setting  $S = 0$  in (18) and the LMI-based conditions in Theorems 1 and 2 and Corollaries 1 and 2, the reliable fuzzy  $H_\infty$  controller and fuzzy  $H_\infty$  controller can be obtained for the systems (12) and (28) under the condition that the actuator delay  $d(t)$  satisfies  $0 < d(t) \leq \bar{d}$ , respectively.

It can be seen from the LMI-based conditions in Theorem 2 and Corollary 2, both dependent on the matrix variables and the objective scalar  $\gamma$ , which implies that  $\gamma$  can be included as an optimization variable to obtain a lower bound of the guaranteed  $H_\infty$  performance. Based on the different conditions, reliable fuzzy  $H_\infty$  controller and fuzzy  $H_\infty$  controller can be designed with the minimal  $\gamma$  by solving the following convex optimization problems:

$$\begin{aligned}\min \gamma \quad \text{s.t.} \quad & (22)–(24) \\ & \bar{P} > 0, \bar{Q} > 0, \bar{S} > 0, \bar{R} > 0, \bar{\epsilon}_{1ij} > 0, \bar{\epsilon}_{2ij} > 0 \\ & Y_{aj}, \bar{M}, \bar{N}\end{aligned}\quad (35)$$

and

$$\begin{aligned}\min \gamma \quad \text{s.t.} \quad & (32)–(34) \\ & \bar{P} > 0, \bar{Q} > 0, \bar{S} > 0, \bar{R} > 0, Y_{sj}, \bar{M}, \bar{N}.\end{aligned}\quad (36)$$

TABLE I  
QUARTER-CAR MODEL PARAMETERS

| $k_s$    | $k_t$     | $c_s$    | $c_t$    |
|----------|-----------|----------|----------|
| 42720N/m | 101115N/m | 1095Ns/m | 14.6Ns/m |

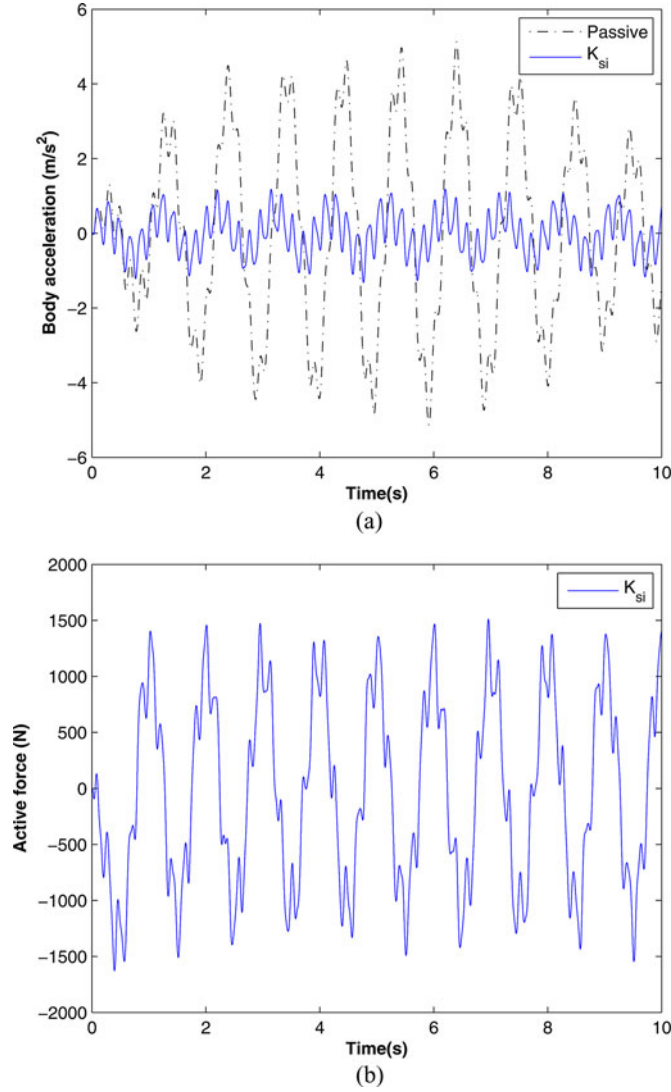


Fig. 3. (a) Responses of body vertical accelerations. (b) Response of active force.

## V. SIMULATION RESULTS

A quarter-vehicle active suspension system is exploited to demonstrate the effectiveness of the proposed approach in this section. The quarter-vehicle suspension model parameters in Table I are used in this study.

The sprung mass  $m_s(t)$  is assumed to set as the range [873 kg, 1073 kg] and the unsprung mass  $m_u(t)$  to [104 kg, 124 kg]. In this study, the maximum allowable suspension stroke is set as  $z_{\max} = 0.1$  m with  $\rho = 1$ . For the actuator delay, i.e.,  $d(t) = 5 + 5 \sin(\frac{1}{50})$  ms, satisfying  $\bar{d} = 10$  ms and  $\mu = 0.1$ , we consider fuzzy  $H_\infty$  controller design for the uncertain active suspension systems in (28). By using the convex

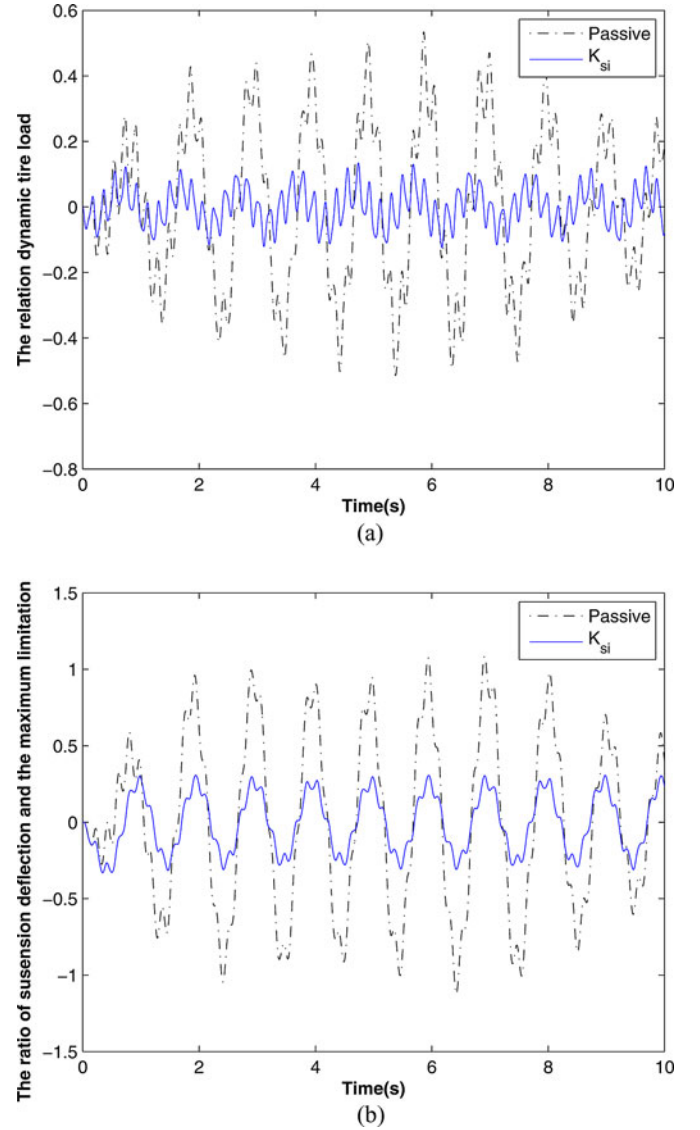


Fig. 4. (a) Responses of suspension deflection constraint. (b) Responses of tire stroke constraint.

optimization in (36), it is found that the minimum guaranteed closed-loop  $H_\infty$  performance index  $\gamma_{\min}$  is 5.3011, and the fuzzy controller gain matrices

$$K_{si} = 10^4 \times [-3.3260 \quad 5.6998 \quad -2.5167 \quad 0.2824] \quad (37)$$

where  $i = 1, 2, 3$ , and 4.

It is expected that the desired fuzzy  $H_\infty$  controller in (27) with the parameters in (37) can be designed, such that 1) the sprung mass acceleration  $z_1(t)$  is as small as possible; 2) the suspension deflection is less than the maximum allowable suspension stroke, i.e.,  $z_{\max} = 0.1$  m, which means that  $x_1(t)/z_{\max}$  is less than 1; and 3) the relation dynamic tire load, i.e.,  $k_t x_2(t)/(m_s(t) + m_u(t))g < 1$ . We first consider the test road disturbance as

$$z_r(t) = 0.0254 \sin 2\pi t + 0.005 \sin 10.5\pi t + 0.001 \sin 21.5\pi t(m). \quad (38)$$

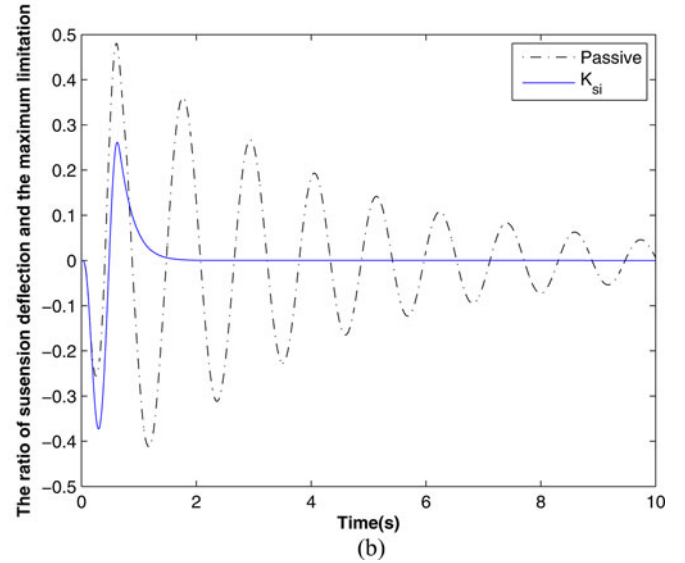
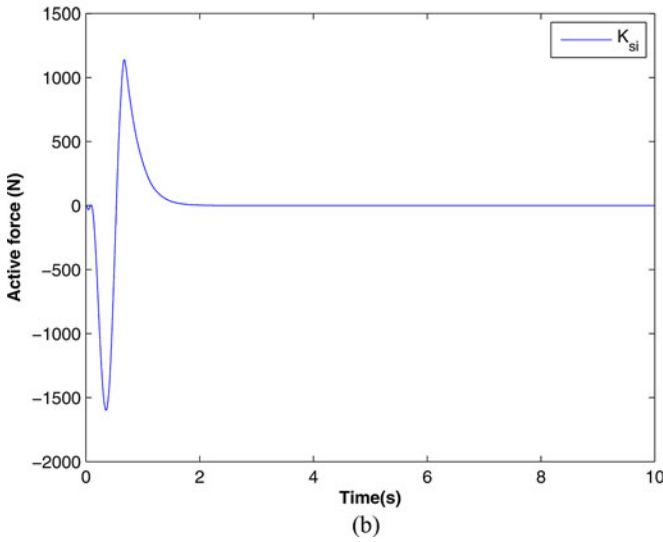
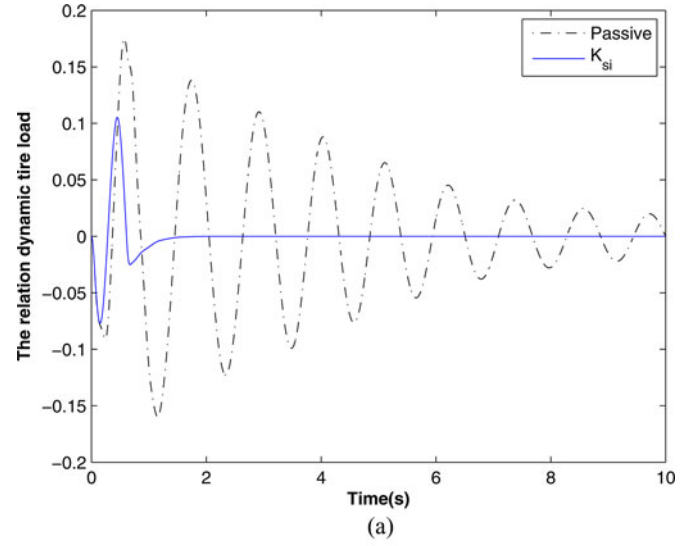
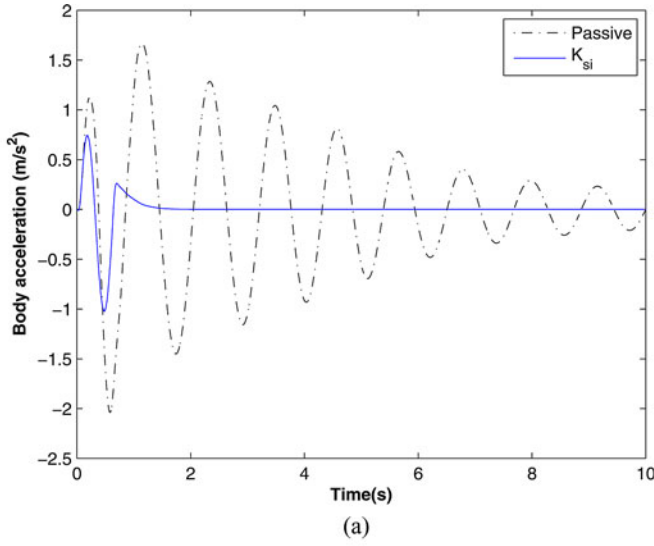


Fig. 5. (a) Responses of body vertical accelerations. (b) Response of active force.

Fig. 6. (a) Responses of suspension deflection constraint. (b) Responses of tire stroke constraint.

According to [47], the road disturbance has the similar frequency as the car body resonance frequency (1 Hz) under the condition that high-frequency disturbance is added to simulate the rough road surface. In order to carry out the simulation for the fuzzy  $H_\infty$  controller as in (28), the variational sprung mass  $m_s(t)$  and the variational unsprung mass  $m_u(t)$  are set as  $m_s(t) = 973 + 100 \sin(t)$  kg and  $m_u(t) = 114 + 10 \cos(t)$  kg, respectively, to derive the fuzzy membership functional  $h_i(\xi(t))$  ( $i = 1, 2, 3, 4$ ). By the use of the fuzzy  $H_\infty$  controller in (27) with the parameters in (37), we derive the corresponding closed-loop fuzzy system. Fig. 3 depicts the responses of body vertical accelerations and the actuator force for the open-loop (e.g., passive) and closed-loop (e.g., active) systems. Fig. 4 demonstrates the responses of suspension stroke and tire deflection constraint for both the passive and active systems. It is observed, from Fig. 3, that the proposed fuzzy  $H_\infty$  control strategy reduces the sprung mass acceleration significantly in comparison with the passive suspension under the same road

disturbance. The designed fuzzy  $H_\infty$  controller can achieve the less value of the maximum body acceleration for the active suspension system than the passive system, and the passenger acceleration in the active suspension system is reduced significantly, which guarantees better ride comfort. In addition, it can be seen that, from Fig. 4, the suspension deflection constraint, i.e.,  $x_1(t)/z_{\max} < 1$ , and the relation dynamic tire load constraint, i.e.,  $k_t x_2(t)/(m_s(t) + m_u(t))g < 1$ , are guaranteed, which implies that the road holding capability is ensured by the desired fuzzy controller. These two figures confirm that the designed standard state-feedback fuzzy  $H_\infty$  controller can achieve better ride comfort and road handling and guarantee constraint suspension deflection for the active suspension system.

To further evaluate the effectiveness of the proposed fuzzy  $H_\infty$  controller design strategy with actuator delays, the road disturbance is taken into account. In the context of active suspension performance, the road disturbance can be generally assumed as discrete events of relatively short duration and high

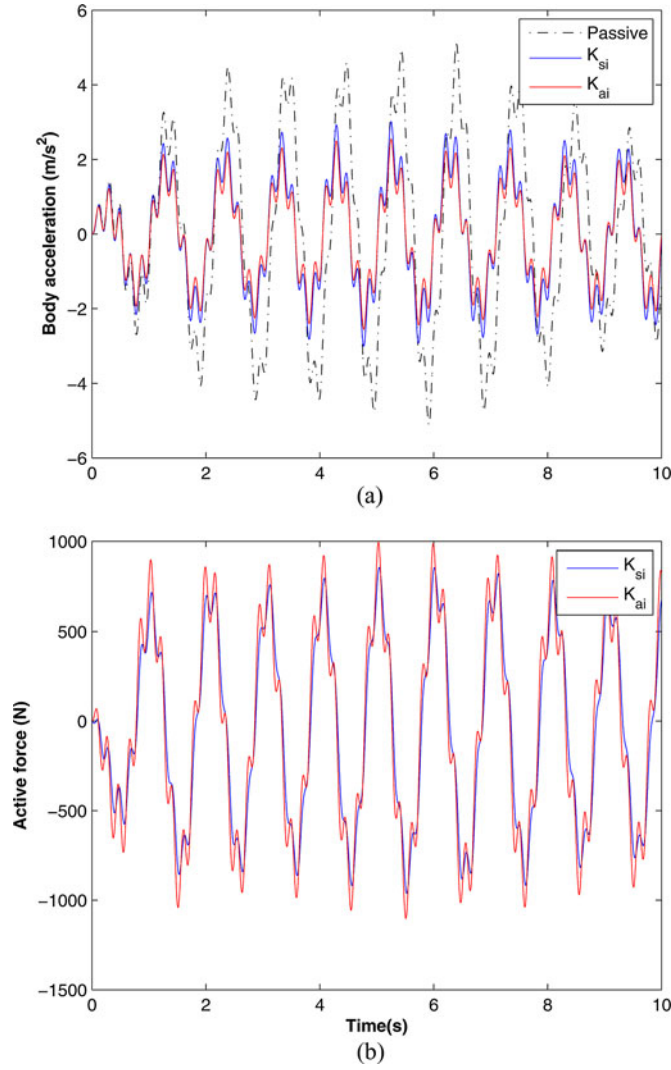


Fig. 7. (a) Responses of body vertical accelerations. (b) Response of active force.

intensity, which is caused by, e.g., a pronounced bump or pot-hole on an otherwise smooth road surface. The road surface is represented as

$$z_r(t) = \begin{cases} \frac{A}{2} \left( 1 - \cos \left( \frac{2\pi V}{L} t \right) \right), & \text{if } 0 \leq t \leq \frac{L}{V} \\ 0, & \text{if } t > \frac{L}{V} \end{cases} \quad (39)$$

where  $A$  and  $L$  are the height and the length of the bump. Assume  $A = 50$  mm,  $L = 6$  m, and the vehicle forward velocity as  $V = 35$  (km/h). Fig. 5 illustrates the responses to body vertical accelerations and the actuator force. Fig. 6 presents the responses to suspension stroke and tire deflection constraint for the passive and active systems under the introduced road disturbance, respectively. The simulation results convincingly demonstrate that the fuzzy  $H_\infty$  controller offers better suspension performance than the open-loop suspension system.

The effectiveness and advantages of the proposed reliable fuzzy  $H_\infty$  controller design for active suspension systems with actuator delay and fault will be demonstrated in what follows.

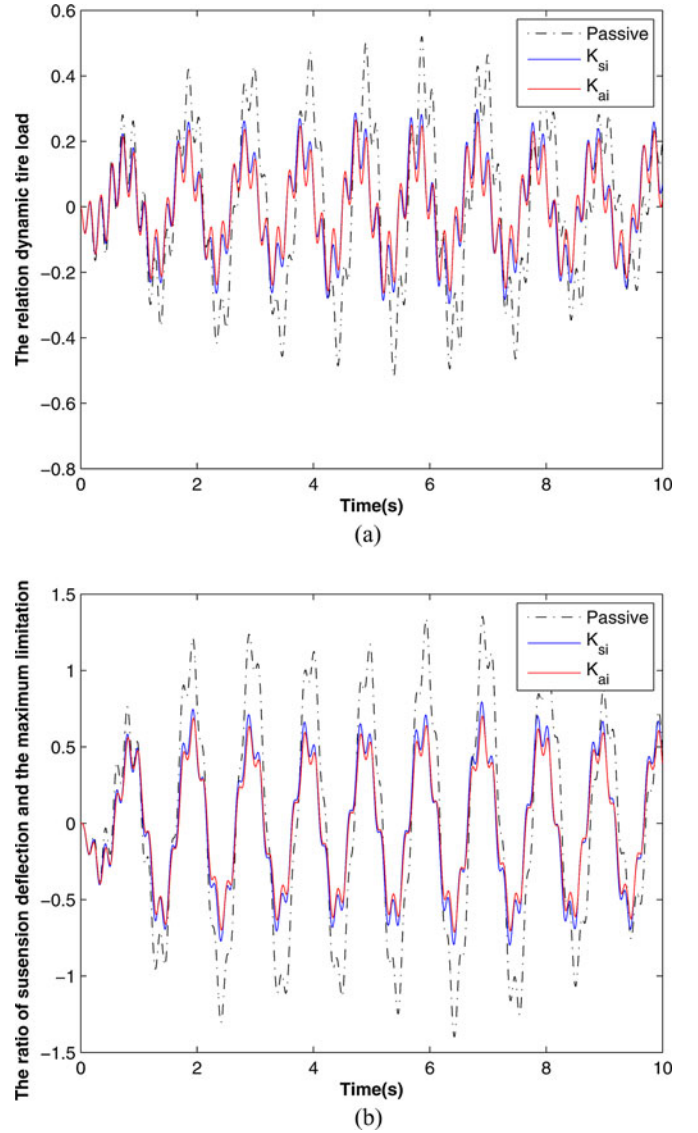


Fig. 8. (a) Responses of suspension deflection constraint. (b) Responses of tire stroke constraint.

The parameters' notations in the fuzzy  $H_\infty$  controller design in the previous section are applied here as well. It is assumed that there exists the following actuator fault, namely,  $\tilde{m}_a = 0.1$  and  $\hat{m}_a = 0.5$ , which implies  $M_{a0} = 0.3$  and  $J_a = 0.2$ . Based on the convex optimization that is presented in (35), we can obtain the minimum guaranteed closed-loop  $H_\infty$  performance index  $\gamma_{\min} = 28.6991$  and the reliable fuzzy controller gain matrices

$$\begin{aligned} K_{a1} &= 10^4 \times [4.1910 \quad -0.9700 \quad -2.5381 \quad 0.5713] \\ K_{a2} &= 10^4 \times [4.1916 \quad -0.9829 \quad -2.5381 \quad 0.5711] \\ K_{a3} &= 10^4 \times [4.1964 \quad -0.9751 \quad -2.5382 \quad 0.5706] \\ K_{a4} &= 10^4 \times [4.2149 \quad -0.9439 \quad -2.5388 \quad 0.5701]. \end{aligned} \quad (40)$$

For the two kinds of road disturbances, namely, the first case road disturbance is shown in (38) and the second case road disturbance as given in (39). In Figs. 7–10, the responses to the open- and closed-loop systems with the actuator delay and

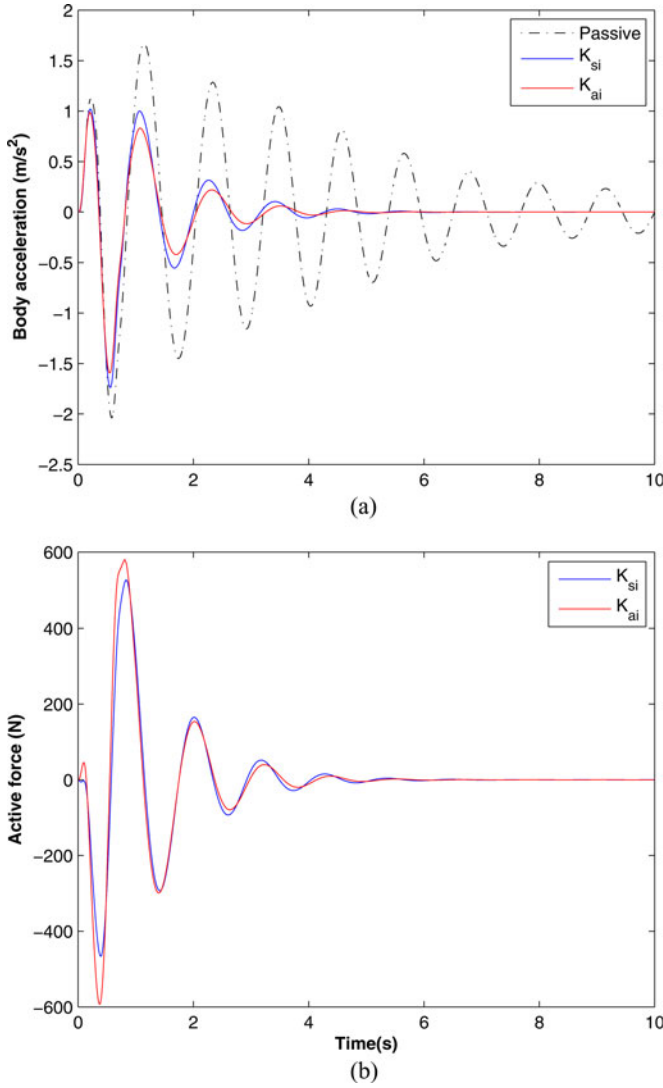


Fig. 9. (a) Responses of body vertical accelerations. (b) Response of active force.

fault via the standard fuzzy  $H_\infty$  controller  $K_{si}$  and reliable controller  $K_{ai}$  ( $i = 1, 2, 3, 4$ ) are based on the two different types of road disturbances. These figures show that the less value of the maximum body acceleration is achieved for the active suspension system, the suspension deflection constraint, i.e.,  $x_1(t)/z_{\max} < 1$ , is guaranteed, and the relation dynamic tire load  $k_t x_2(t)/(m_s(t) + m_u(t))g$  is less than 1, in comparison with the passive suspension system, by utilizing the standard fuzzy  $H_\infty$  controller  $K_{si}$  and reliable controller  $K_{ai}$  ( $i = 1, 2, 3, 4$ ) for three different types of road disturbances, respectively. However, it can be observed, from Figs. 7 and 2, that the reliable fuzzy  $H_\infty$  controller achieves less value of the maximum body acceleration than the standard  $H_\infty$  controller for the active suspension system with actuator delay and fault. From Figs. 8 and 10, it can be seen that  $K_{ai}$  ( $i = 1, 2, 3, 4$ ) is capable to provide a much more steady control force under the fault condition than the conventional controller  $K_{si}$  ( $i = 1, 2, 3, 4$ ).

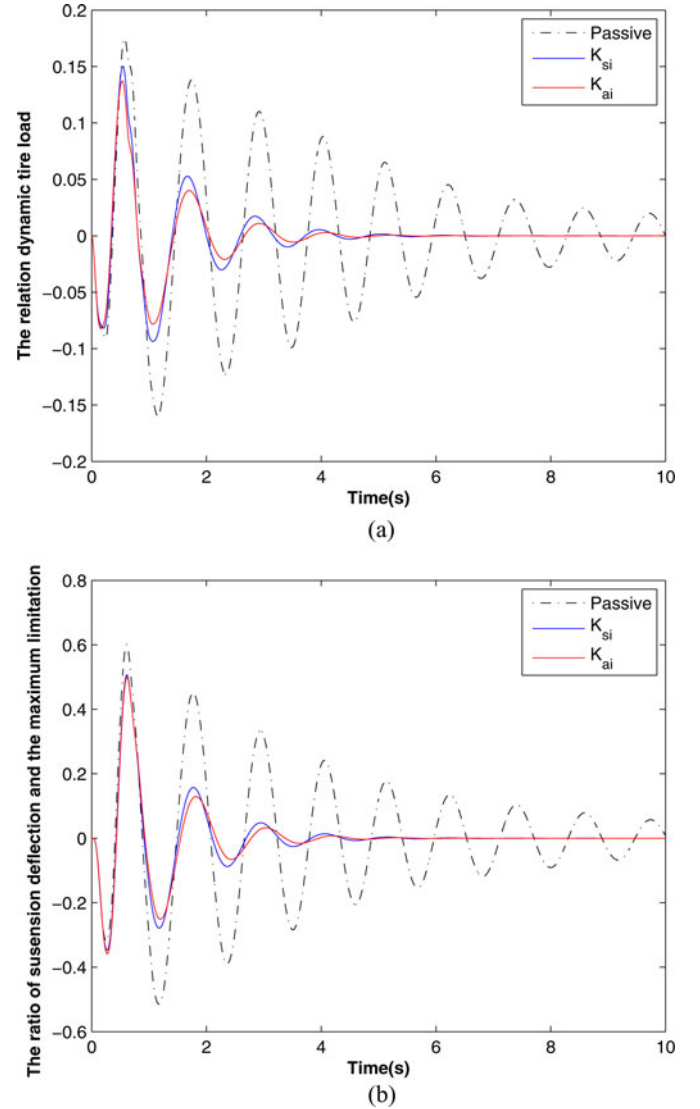


Fig. 10. (a) Responses of suspension deflection constraint. (b) Responses of tire stroke constraint.

To further evaluate the suspension system performance under different fuzzy controllers  $K_{si}$  and  $K_{ai}$  ( $i = 1, 2, 3, 4$ ), the RMS values of the body acceleration are exploited to demonstrate its advantages. The road disturbances can also be generally assumed as random vibrations, which are consistent and typically specified as the random process with a given ground displacement power spectral density (PSD) of

$$G_q(n) = G_q(n_0) \left( \frac{n}{n_0} \right)^{-c} \quad (41)$$

where  $n_0$  denotes the spatial frequency, and  $n_0$  is the reference spatial frequency of  $n_0 = 0.1$  (1/m);  $G_q(n_0)$  stands for the road roughness coefficient; and  $c = 2$  is the road roughness constant. Related to the time frequency  $f$ , we have  $f = nV$  with  $V$  being the vehicle forward velocity. Based on (41), we can obtain the PSD ground displacement

$$G_q(f) = G_q(n_0) n_0^{-2} \frac{V}{f^2}. \quad (42)$$

TABLE II  
RMS BODY ACCELERATION

| Parameter                                    | Passive systems | Fuzzy Controller | Reliable Fuzzy Controller |
|----------------------------------------------|-----------------|------------------|---------------------------|
| $G_q(n_0) = 16 \times 10^{-6} \text{ m}^3$   | 0.0081          | 0.0046           | 0.0041                    |
| $G_q(n_0) = 64 \times 10^{-6} \text{ m}^3$   | 0.0152          | 0.0092           | 0.0083                    |
| $G_q(n_0) = 256 \times 10^{-6} \text{ m}^3$  | 0.0284          | 0.0183           | 0.0166                    |
| $G_q(n_0) = 1024 \times 10^{-6} \text{ m}^3$ | 0.0644          | 0.0387           | 0.0351                    |

TABLE III  
RMS SUSPENSION STROKE

| Parameter                                    | Passive systems         | Fuzzy Controller        | Reliable Fuzzy Controller |
|----------------------------------------------|-------------------------|-------------------------|---------------------------|
| $G_q(n_0) = 16 \times 10^{-6} \text{ m}^3$   | $1.7635 \times 10^{-4}$ | $9.7651 \times 10^{-5}$ | $9.5584 \times 10^{-5}$   |
| $G_q(n_0) = 64 \times 10^{-6} \text{ m}^3$   | $3.3536 \times 10^{-4}$ | $1.9626 \times 10^{-4}$ | $1.9057 \times 10^{-4}$   |
| $G_q(n_0) = 256 \times 10^{-6} \text{ m}^3$  | $6.2909 \times 10^{-4}$ | $3.9088 \times 10^{-4}$ | $3.8283 \times 10^{-4}$   |
| $G_q(n_0) = 1024 \times 10^{-6} \text{ m}^3$ | 0.0014                  | $8.2616 \times 10^{-4}$ | $8.0992 \times 10^{-4}$   |

TABLE IV  
RMS RELATIVE DYNAMICS TIRE LOAD

| Parameter                                    | Passive systems         | Fuzzy Controller        | Reliable Fuzzy Controller |
|----------------------------------------------|-------------------------|-------------------------|---------------------------|
| $G_q(n_0) = 16 \times 10^{-6} \text{ m}^3$   | $8.3596 \times 10^{-4}$ | $5.2554 \times 10^{-4}$ | $4.9612 \times 10^{-4}$   |
| $G_q(n_0) = 64 \times 10^{-6} \text{ m}^3$   | 0.0016                  | 0.0010                  | $9.9561 \times 10^{-4}$   |
| $G_q(n_0) = 256 \times 10^{-6} \text{ m}^3$  | 0.0030                  | 0.0021                  | 0.0020                    |
| $G_q(n_0) = 1024 \times 10^{-6} \text{ m}^3$ | 0.0067                  | 0.0044                  | 0.0042                    |

Accordingly, the PSD ground velocity is given by

$$G_{\dot{q}}(f) = (2\pi f)^2 G_q(f) = 4\pi G_q(n_0) n_0^2 V \quad (43)$$

which is only related to the vehicle forward velocity. When the vehicle forward velocity is fixed, the ground velocity can be viewed as a white-noise signal. We choose the four difference road roughness, i.e.,  $G_q(n_0) = 16 \times 10^{-6}$ ,  $64 \times 10^{-6}$ ,  $256 \times 10^{-6}$ , and  $1024 \times 10^{-6} \text{ m}^3$ , which correspond to a B (Good), C (Average), D (Poor), and E (Very Poor) grade for the vehicle forward velocity, i.e.,  $V = 35 \text{ km/h}$ , respectively

The RMS values are strictly related to the ride comfort, which are often used to quantify the amount of acceleration transmitted to the vehicle body. The RMS value of variable  $x(t)$  is calculated as  $\text{RMS}_x = \sqrt{(1/T) \int_0^T x^T(t)x(t)dt}$ . In our study, we choose  $T = 100 \text{ s}$  to calculate the RMS values of the body acceleration, suspension stroke, and relative dynamics tire load for different road roughness coefficient  $G_q(n_0)$ , which are listed in Tables II–IV by the use of the fuzzy controller  $K_{si}$  and reliable fuzzy controller  $K_{ai}$ , respectively. It can be observed that these tables indicate that the improvement in ride comfort and the satisfaction of hard constraints can be achieved for the different load conditions by the use of the reliable fuzzy controller  $K_{ai}$  compared with the fuzzy controller  $K_{si}$  for the uncertain suspension systems with actuator delay and fault.

## VI. CONCLUSION

In this paper, the problem of reliable fuzzy  $H_\infty$  control for active suspension systems with actuator delay and fault has been investigated. The sprung and unsprung mass variations, the actuator delay and fault, and the suspension performance have all been taken into account to construct the T–S fuzzy system for the control design objective. Based on the PDC scheme

and stability theory, the reliable fuzzy  $H_\infty$  performance analysis condition has been derived for the proposed T–S fuzzy system, presenting the active suspension system with uncertainty. Then, the reliable fuzzy  $H_\infty$  controller has been designed such that the resulting closed-loop T–S fuzzy system is asymptotically stable with  $H_\infty$  performance and simultaneously satisfies the constraint suspension performance. A quarter-vehicle suspension model has been used to validate the effectiveness of the proposed design method. Simulation results have clearly demonstrated that the designed reliable fuzzy controller has the capability of guaranteeing a better suspension performance under sprung and unsprung mass variations, actuator delay, and fault.

## ACKNOWLEDGMENT

The authors would like to thank the Associate Editor and the anonymous referees for their constructive comments and valuable suggestions, which helped improve the quality of this paper.

## REFERENCES

- [1] D. Hrovat, "Survey of advanced suspension developments and related optimal control applications," *Automatica*, vol. 33, no. 10, pp. 1781–1817, 1997.
- [2] H. Gao, W. Sun, and P. Shi, "Robust sampled-data  $H_\infty$  control for vehicle active suspension systems," *IEEE Trans. Control Syst. Technol.*, vol. 18, no. 1, pp. 238–245, Jan. 2010.
- [3] J. Cao, H. Liu, P. Li, and D. Brown, "State of the art in vehicle active suspension adaptive control systems based on intelligent methodologies," *IEEE Trans. Intell. Transport. Syst.*, vol. 9, no. 3, pp. 392–405, Sep. 2008.
- [4] M. Zapateiro, F. Pozo, H. Karimi, and N. Luo, "Semiactive control methodologies for suspension control with magnetorheological dampers," *IEEE/ASME Trans. Mechatronics*, doi:10.1109/TMECH.2011.2107331.

- [5] M. Yamashita, K. Fujimori, K. Hayakawa, and H. Kimura, "Application of  $H_\infty$  control to active suspension systems," *Automatica*, vol. 30, no. 11, pp. 1717–1729, 1994.
- [6] A. Alleyne and J. Hedrick, "Nonlinear adaptive control of active suspensions," *IEEE Trans. Control Syst. Technol.*, vol. 3, no. 1, pp. 94–101, Mar. 1995.
- [7] H. Du, K. Y. Sze, and J. Lam, "Semi-active  $H_\infty$  control of vehicle suspension with magneto-rheological dampers," *J. Sound Vibrat.*, vol. 283, no. 3–5, pp. 981–996, 2005.
- [8] N. Al-Holou, T. Lahdhiri, D. Joo, J. Weaver, and F. Al-Abbas, "Sliding mode neural network inference fuzzy logic control for active suspension systems," *IEEE Trans. Fuzzy Syst.*, vol. 10, no. 2, pp. 234–246, Apr. 2002.
- [9] C. Sivrioglu and I. Cansever, "LPV gain-scheduling controller design for a non-linear quarter-vehicle active suspension system," *Trans. Inst. Meas. Control*, vol. 31, no. 1, pp. 71–95, Feb. 2009.
- [10] M. ElMadany and Z. Abduljabbar, "Linear quadratic Gaussian control of a quarter-car suspension," *Veh. Syst. Dyn.*, vol. 32, no. 6, pp. 479–497, 1999. DOI: 10.1109/TMECH.2011.2107331.
- [11] I. Fialho and G. Balas, "Road adaptive active suspension design using linear parameter-varying gain-scheduling," *IEEE Trans. Control Syst. Technol.*, vol. 10, no. 1, pp. 43–54, Jan. 2002.
- [12] H. Chen and K. Guo, "Constrained  $H_\infty$  control of active suspensions: An LMI approach," *IEEE Trans. Control Syst. Technol.*, vol. 13, no. 3, pp. 412–421, May 2005.
- [13] H. Karimi, M. Zapateiro, and N. Luo, "A linear matrix inequality approach to  $H_\infty$  control of vehicle engine–body vibration systems with a time-varying actuator delay," in *Proc. Inst. Mech. Eng., Part I: J. Syst. Control Eng.*, vol. 222, no. 8, pp. 883–899, 2008.
- [14] G. Yang, S. Zhang, J. Lam, and J. Wang, "Reliable control using redundant controllers," *IEEE Trans. Automat. Control*, vol. 43, no. 11, pp. 1588–1593, Nov. 2002.
- [15] P. Shi, E. Boukas, S. Nguang, and X. Guo, "Robust disturbance attenuation for discrete-time active fault tolerant control systems with uncertainties," *Optimal Control Appl. Methods*, vol. 24, no. 2, pp. 85–101, 2003.
- [16] Y. Guo, B. Jiang, and P. Shi, "Delay-dependent adaptive reconfiguration control in the presence of input saturation and actuator faults," *Int. J. Innovat. Comput., Inf. Control*, vol. 6, no. 4, pp. 1873–1882, 2010.
- [17] Z. Wang, G. Wei, and G. Feng, "Reliable  $H_\infty$  control for discrete-time piecewise linear systems with infinite distributed delays," *Automatica*, vol. 45, no. 12, pp. 2991–2994, 2009.
- [18] Z. Mao, B. Jiang, and P. Shi, "Fault-tolerant control for a class of nonlinear sampled-data systems via a Euler approximate observer," *Automatica*, vol. 46, no. 11, pp. 1852–1859, 2010.
- [19] L. Meng and B. Jiang, "Robust active fault-tolerant control for a class of uncertain nonlinear systems with actuator faults," *Int. J. Innovat. Comput., Inf. Control*, vol. 6, no. 6, pp. 2637–2644, 2010.
- [20] G. Zhou, P. Chen, and T. Engang, "Reliable control for a class of discrete-time state-delayed nonlinear systems with stochastic actuators failures," *ICIC Expr. Lett.*, vol. 4, no. 6, pp. 2475–2480, 2010.
- [21] G. Yang, J. Wang, and Y. Soh, "Reliable  $H_\infty$  controller design for linear systems," *Automatica*, vol. 37, no. 5, pp. 717–725, 2001.
- [22] Y. Zhao, L. Zhao, and H. Gao, "Vibration control of seat suspension using  $H_\infty$  reliable control," *J. Vibrat. Control*, vol. 16, no. 12, pp. 1859–1879, 2010.
- [23] Y. S. Moon, P. G. Park, and W. H. Kwon, "Robust stabilization of uncertain input-delayed systems using reduction method," *Automatica*, vol. 37, pp. 307–312, 2001.
- [24] S. Xu and J. Lam, "Robust  $H_\infty$  control for uncertain discrete-time-delay fuzzy systems via output feedback controllers," *IEEE Trans. Fuzzy Syst.*, vol. 13, no. 1, pp. 82–93, Feb. 2005.
- [25] C. Lin, Z. Wang, and F. Yang, "Observer-based networked control for continuous-time systems with random sensor delays," *Automatica*, vol. 45, no. 2, pp. 578–584, 2009.
- [26] B. Chen, X. Liu, S. Tong, and C. Lin, "Observer-based stabilization of T-S fuzzy systems with input delay," *IEEE Trans. Fuzzy Syst.*, vol. 16, no. 3, pp. 652–663, Jun. 2008.
- [27] X. M. Zhang, M. Wu, J. H. She, and Y. He, "Delay-dependent stabilization of linear systems with time-varying state and input delays," *Automatica*, vol. 41, pp. 1405–1412, 2005.
- [28] H. Gao, J. Lam, and C. Wang, "Multi-objective control of vehicle active suspension systems via load-dependent controllers," *J. Sound Vibrat.*, vol. 290, no. 3–5, pp. 654–675, 2006.
- [29] H. Du, N. Zhang, and J. Lam, "Parameter-dependent input-delayed control of uncertain vehicle suspensions," *J. Sound Vibrat.*, vol. 317, no. 3–5, pp. 537–556, 2008.
- [30] L. Zadeh, "Fuzzy sets," *Inf. Control*, vol. 8, no. 3, pp. 338–353, 1965.
- [31] M. Sugeno, "An introductory survey of fuzzy control," *Inf. Sci.*, vol. 36, nos. 1–2, pp. 59–83, 1985.
- [32] K. Tanaka and H. Wang, *Fuzzy Control Systems Design and Analysis: A Linear Matrix Inequality Approach*. New York: Wiley-Interscience, 2001.
- [33] G. Feng, "A survey on analysis and design of model-based fuzzy control systems," *IEEE Trans. Fuzzy Syst.*, vol. 14, no. 5, pp. 676–697, Oct. 2006.
- [34] C. Lin, G. Wang, and T. Lee, *LMI Approach to Analysis and Control of Takagi-Sugeno Fuzzy Systems With Time Delay*. New York: Springer-Verlag, 2007.
- [35] H. Lam and M. Narimani, "Quadratic-stability analysis of fuzzy-model-based control systems using staircase membership functions," *IEEE Trans. Fuzzy Syst.*, vol. 18, no. 1, pp. 125–137, Feb. 2010.
- [36] S. Nguang and P. Shi, " $H_\infty$  fuzzy output feedback control design for nonlinear systems: An LMI approach," *IEEE Trans. Fuzzy Syst.*, vol. 11, no. 3, pp. 331–340, Jun. 2003.
- [37] B. Zhang and S. Xu, "Delay-dependent robust  $H_\infty$  control for uncertain discrete-time fuzzy systems with time-varying delays," *IEEE Trans. Fuzzy Syst.*, vol. 17, no. 4, pp. 809–823, Aug. 2009.
- [38] J. Zhang, P. Shi, and Y. Xia, "Robust adaptive sliding-mode control for fuzzy systems with mismatched uncertainties," *IEEE Trans. Fuzzy Syst.*, vol. 18, no. 4, pp. 700–711, Aug. 2010.
- [39] B. Jiang, Z. Mao, and P. Shi, " $H_\infty$ -filter design for a class of networked control systems via T-S fuzzy-model approach," *IEEE Trans. Fuzzy Syst.*, vol. 18, no. 1, pp. 201–208, Feb. 2010.
- [40] H. Dong, Z. Wang, D. Ho, and H. Gao, "Robust  $H_\infty$  fuzzy output-feedback control with multiple probabilistic delays and multiple missing measurements," *IEEE Trans. Fuzzy Syst.*, vol. 18, no. 4, pp. 712–725, Aug. 2010.
- [41] B. Chen and X. Liu, "Reliable control design of fuzzy dynamic systems with time-varying delay," *Fuzzy Sets and Syst.*, vol. 146, no. 3, pp. 349–374, 2004.
- [42] H. Wu and H. Zhang, "Reliable  $H_\infty$  fuzzy control for continuous-time nonlinear systems with actuator failures," *IEEE Trans. Fuzzy Syst.*, vol. 14, no. 5, pp. 609–618, Oct. 2006.
- [43] Y. Wang, D. Zhou, and L. Liu, "Reliable memory feedback design for a class of nonlinear fuzzy systems with time-varying delay," *Int. J. Autom. Comput.*, vol. 4, no. 2, pp. 169–176, 2007.
- [44] S. Nguang, P. Shi, and S. Ding, "Fault detection for uncertain fuzzy systems: An LMI approach," *IEEE Trans. Fuzzy Syst.*, vol. 15, no. 6, pp. 1251–1262, Dec. 2007.
- [45] J. Cao, P. Li, and H. Liu, "An interval fuzzy controller for vehicle active suspension systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 11, no. 4, pp. 885–895, Dec. 2010.
- [46] S. Huang and W. Lin, "Adaptive fuzzy controller with sliding surface for vehicle suspension control," *IEEE Trans. Fuzzy Syst.*, vol. 11, no. 4, pp. 550–559, Aug. 2003.
- [47] H. Du and N. Zhang, "Fuzzy control for nonlinear uncertain electrohydraulic active suspensions with input constraint," *IEEE Trans. Fuzzy Syst.*, vol. 17, no. 2, pp. 343–356, Apr. 2009.
- [48] Y. He, Q. Wang, C. Lin, and M. Wu, "Delay-range-dependent stability for systems with time-varying delay," *Automatica*, vol. 43, no. 2, pp. 371–376, 2007.
- [49] H. Shao, "New delay-dependent stability criteria for systems with interval delay," *Automatica*, vol. 45, no. 3, pp. 744–749, 2009.



**Hongyi Li** (S'11) received the B.S. and M.S. degrees in mathematics from Bohai University, Jinzhou, China, in 2006 and 2009, respectively. He is currently working toward the Ph.D. degree with the international collaboration program, University of Portsmouth, Portsmouth, U.K., and Harbin Institute of Technology, Harbin, China.

His research interest includes fuzzy control, robust control, and their applications in suspension systems.



**Honghai Liu** (M'02–SM'06) received the Ph.D. degree in intelligent robotics from Kings College, University of London, London, U.K., in 2003.

He is currently a Professor of intelligent systems with the University of Portsmouth, Portsmouth, U.K. He previously held research appointments with the University of London and the University of Aberdeen, Aberdeen, U.K., with project leader appointments in the large-scale industrial control and system integration industry. His research interests include approximate computation, pattern recognition, intelligent video analytics, cognitive robotics, and their practical applications, with an emphasis on approaches which could make contributions to the intelligent connection of perception to action using contextual information.

Dr. Liu is a Fellow of the Institution of Engineering and Technology.

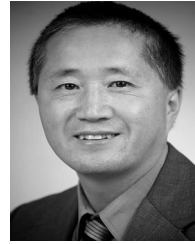


**Huijun Gao** (SM'09) received the Ph.D. degree in control science and engineering from the Harbin Institute of Technology, Harbin, China, in 2005.

He was a Research Associate with the Department of Mechanical Engineering, The University of Hong Kong, Hong Kong, from November 2003 to August 2004. From October 2005 to October 2007, he carried out his postdoctoral research with the Department of Electrical and Computer Engineering, University of Alberta, Edmonton, AB, Canada. Since November 2004, he has been with the Harbin Institute of Technology,

where he is currently a Professor and Director of the Research Institute of Intelligent Control and Systems. His research interests include network-based control, robust control/filter theory, time-delay systems, and their engineering applications.

Dr. Gao is an Associate Editor of *Automatica*; the IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS; the IEEE TRANSACTIONS ON SYSTEMS MAN AND CYBERNETICS PART B: CYBERNETICS; the IEEE TRANSACTIONS ON FUZZY SYSTEMS; the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS-I; and the IEEE TRANSACTIONS ON CONTROL SYSTEMS TECHNOLOGY.



**Peng Shi** (SM'97) received the B.Sc. degree in mathematics from the Harbin Institute of Technology, Harbin, China, in 1982, the M.E. degree in systems engineering from Harbin Engineering University, in 1985, the Ph.D. degree in electrical engineering from the University of Newcastle, Newcastle, Australia, in 1994, and the Ph.D. degree in mathematics from the University of South Australia, Adelaide, Australia, in 1998. He received the Doctor of Science degree from the University of Glamorgan, Pontypridd, U.K., in 2006.

He was a Lecturer with Heilongjiang University, Harbin, in 1985–1989 and with the University of South Australia in 1997–1999 and a Senior Scientist with the Defence Science and Technology Organization, Department of Defence, Australia, during 1999–2005. He joined the University of Glamorgan as a Professor in 2004. He has also been a Professor with Victoria University, Melbourne, Australia, since 2008. His research interests include control system design, fault detection techniques, Markov decision processes, and operational research. He has published a number of papers in these areas. In addition, he is a coauthor of the three research monographs: *Analysis and Synthesis of Systems with Time-Delays* (Berlin, Germany: Springer, 2009), *Fuzzy Control and Filtering Design for Uncertain Fuzzy Systems* (Berlin, Germany: Springer, 2006), and *Methodologies for Control of Jump Time-Delay Systems* (Boston, MA: Kluwer, 2003).

Dr. Shi currently serves as the Editor-in-Chief of *International Journal of Innovative Computing, Information and Control*. He is also an Advisory Board Member, Associate Editor, and Editorial Board Member for a number of international journals, including the IEEE TRANSACTIONS ON AUTOMATIC CONTROL; the IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS-PART B; the IEEE TRANSACTIONS ON FUZZY SYSTEMS; *Information Sciences*; and the *International Journal of Systems Science*. He received the Most Cited Paper Award of Signal Processing in 2009. He is a Fellow of the Institute of Engineering and Technology (U.K.) and a Fellow of the Institute of Mathematics and its Applications (U.K.).